



Digital twins assisted for fault diagnosis of rotating machinery: a review

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ABSTRACT

As an emerging and efficient diagnostic tool, digital twin (DT) technology has been widely recognized and applied in rotating machinery fault diagnosis, providing unprecedented support for the reliability and efficiency of industrial systems. In this review, the main research methods of DT, construction methods of DT models, twin data types, and specific applications in rotating machinery are comprehensively reviewed. Moreover, the latest research on DT-assisted fault diagnosis of rotating machinery is systematically summarized. Additionally, the key challenges in current DT applications, such as building high-fidelity digital models, meeting real-time condition monitoring and diagnosis requirements, addressing model lightweight and generalization issues, and integrating physical and data-driven models, are identified and discussed in depth. Looking ahead, advancements in multimodal data fusion and model lightweighting are expected to drive further progress in DT model development. Meanwhile, the integration of edge computing with cloud collaboration will markedly improve the real-time capabilities and robustness of diagnostic systems. Furthermore, the continued development of data-driven neural network architectures is anticipated to improve model adaptability under complex conditions and extreme fault scenarios. DT technology is progressively enabling a shift in rotating machinery fault diagnosis from reactive maintenance toward proactive health management. This article aims to enable researchers to quickly and comprehensively grasp the current state of research, methods, basic theories, existing challenges, and key directions for future exploration, thereby promoting the widespread integration of DT technology with fault diagnosis of rotating machinery and facilitating the development of universally applicable methods across different devices and scenarios.

1. Introduction

Rotating machinery is core equipment of modern industrial systems and is utilized extensively in energy, aerospace, transportation and other industries. The operating condition of such machinery is directly linked to production efficiency, equipment lifespan, and operational safety. Condition monitoring and fault diagnosis have been proven to play a critical role in ensuring system safety, improving equipment reliability, and reducing maintenance costs and property damage [1]. Traditional fault diagnosis methods rely on substantial fault data, signal processing techniques, and the experiential knowledge of domain experts [2]. However, it is very difficult to collect equipment fault data under actual operating conditions, and the available data are often unbalanced and incomplete [3]. These limitations necessitate the development of more sophisticated and adaptive fault diagnosis methods to effectively overcome inherent challenges, thereby enhancing diagnostic accuracy and operational reliability.

The recent advancements in sensing technology, artificial intelligence, the Industrial Internet of Things (IIoT), and computer technology have precipitated a paradigm shift in intelligent fault diagnosis (IFD), ushering it into an age characterized by big data and digitization. Wang et al. [4] reviewed the research progress of transfer learning (TL)-based domain adaptation techniques for IFD. They summarized the foundational research, current status, theoretical limitations, and unexplored directions, offering valuable insights for future research. Lei et al. [5] reviewed the progress of IFD techniques using machine learning (ML), deep learning (DL), and TL approaches. They further proposed a roadmap for the future development of IFD technologies. Chen et al. [6] reviewed the application of deep transfer learning (DTL) in bearing fault diagnosis from 2016 onward and proposed a novel classification method based on this approach. Sahu et al. [7] analyzed various data-driven fault diagnosis methods for industrial equipment and summarized the advantages and limitations of these approaches. Neupane et al. [8] reviewed the use of various ML methods for detecting and diagnosing

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mechanical faults, highlighting the strengths and limitations of data-driven approaches and suggesting directions for future research. Wang et al. [9] reviewed the research progress of knowledge- and data-driven fault diagnosis methods in industrial settings, summarizing current challenges, future research directions, and potential solutions. Saeed et al. [10] discussed DL-based approaches to intelligent health management and fault diagnosis in industrial machinery, summarizing the state of the art in data processing, model construction, and training optimization. Li et al. [11] summarized the application of DTL methods in industrial fault diagnosis from an algorithmic perspective, and discussed future research challenges and emerging trends in this field. However, IFD methods based on DL are generally developed under the assumption of access to abundant and well-labeled training data. In practical scenarios, however, the collected fault data are often scarce, noisy, or inaccurate, which can substantially degrade diagnostic performance.

Over the past decade, it has been demonstrated that digital twin (DT) has exhibited significant potential in a variety of fields, including manufacturing [12–14], smart cities [15,16], healthcare [17,18], logistics and transportation [19,20], aerospace [21,22], and smart agriculture [23–25]. Recent research progress has highlighted the potential of DT in fault diagnosis. DTs are defined as high-fidelity virtual models that can simulate real-world conditions, thereby enhancing the processes of fault detection and predictive maintenance. The utilization of DTs is increasingly recognized as a transformative approach, with their capacity to replicate physical systems being exploited to enhance fault diagnosis accuracy and operational efficiency. Wang et al. [26] developed a DT-based fault diagnosis method for the planetary gears of a wind turbine to overcome the limitations of traditional approaches, such as data transmission delays, limited visualization in condition monitoring, and insufficient real-time fault feedback. Zhang et al. [27] employed a DT model to create a virtual representation of rolling bearings, enabling the monitoring of their operational conditions and facilitating fault diagnosis even in scenarios with limited fault data. Inspired by DT, Wang et al. [28] integrated numerical simulation models with ML classification for real-time fault diagnosis of bearings. Liu et al. [29] proposed a model-free fault diagnosis method based on a DT system to address the limitations of current fault diagnosis approaches. These approaches heavily rely on historical fault data and specific fault models, which hinder the realization of unmanned operations. Zhou et al. [30] proposed the use of a DT model to generate high-fidelity simulated fault data for rod-fastening rotor systems in gas turbines and aero-engines, addressing uncertainty and challenges in acquiring real-world fault data. This approach ensures the physical accuracy, richness, and diversity of the source domain data. Yu et al. [31] developed a dynamic modeling and parameter identification methodology for machine tools to establish a DT model, based on which they further proposed an IFD framework integrated with the dynamic model. Liu et al. [32] proposed a DT-based fault diagnosis method to overcome the limitations of existing models, including restricted fault type coverage, low similarity between simulated and real data, and inadequate visualization capabilities in condition monitoring. Liu et al. [33] presented a digital multi-twin fusion diagnostic algorithm based on TL to overcome the challenges associated with constructing DT models, including limitations in modeling methods, poor generalization ability, and challenges in cross-domain applications.

The literature review indicates that DT technology facilitates fault diagnosis, enhances data visualization, enriches fault-related data datasets, and addresses the problem of data scarcity in traditional diagnostic methods. These advantages collectively improve diagnostic efficiency and accuracy, and further demonstrate the broad applicability of DT in fault diagnosis.

1.1. Definition of digital twin

In 2003, Michael Grieves from the University of Michigan introduced

the concept of “digital equivalents of physical products” [34], which is widely regarded as the original of the term “digital twin”. However, the term “digital twin” was initially coined by NASA and applied within both NASA and the Air Force Research Laboratory. It was later introduced into industrial production by General Electric [35]. Since then, numerous definitions of “digital twins” have been proposed. Vrabic et al. [36] defined the DT as a digital representation of a physical product or system, consisting of multiple components for real-time monitoring and analysis. Tao et al. [37] proposed a definition of DT as a comprehensive representation of the life-cycle of both physical and virtual entities, encompassing machines, computer models, simulations, and mirror systems. Negri et al. [38] defined DT as a dynamic simulation model that collects data from the physical environment to control the operation of physical devices. Boulos et al. [39] described DT as a virtual representation of a physical entity, featuring a dynamic bidirectional link between the physical entity and its corresponding DT. Lv et al. [40] defined the DT as a dynamic, multidimensional, and multidisciplinary virtual representation of a physical entity, simulating its properties, behaviors, and operating rules.

In summary, DT technology can be defined as the process of creating a high-fidelity virtual model of a physical entity and establishing an accurate, real-time, bidirectional data mapping between the physical and virtual domains. The engineering application and deployment of DT-assisted fault diagnosis are shown in Fig. 1. The architecture is divided into four layers: the physical entity layer, the virtual model layer, the data transmission layer, and the fault diagnosis layer. The physical entity layer consists of industrial equipment and various sensors, and the measured data is transmitted to the data transmission layer through sensors installed on key elements. The virtual model layer builds a digital replica of the device using parameters from the physical entity and performs multi-physics analysis through simulation software. The simulated data is also transmitted to the data transmission layer, which ensures accurate, real-time transmission and secure data storage. After preliminary processing, the data are forwarded to the fault diagnosis layer. The fault diagnosis layer employs neural network approaches, such as DL, to perform feature extraction and fault identification on the fused data.

1.2. Conceptual framework of digital twin

The general framework of the DT system for fault diagnosis is illustrated in Fig. 2. The core idea of the DT method is to construct virtual models that represent physical entities in the real world and to enable accurate, real-time interaction between them through continuous data transmission. Therefore, physical entities, virtual models, and real-time data interaction constitute the three key components of a DT system [34].

Physical entities refer to tangible devices or production systems capable of performing specific tasks, and they constitute the foundation of DTs. Accurately constructing virtual models and using advanced sensors to collect real-time data from physical entities are essential for effective DT implementation. A virtual model is a digital representation of a physical entity that simulates various potential fault scenarios, thereby enabling technicians to better understand fault mechanisms and to develop appropriate maintenance strategies. Data transmission and interaction refer to the bidirectional flow of data and real-time communication between physical entities and virtual models, ensuring that the virtual model accurately reflects the condition of the physical entity and provides feedback to support system control and optimization. By transmitting real data from physical entities to virtual models, the three components form the fundamental framework of DTs and enable the construction of accurate virtual representations. Conversely, transmitting simulation data from virtual models to physical entities enables real-time monitoring, control, and optimization.

The composition and key research areas of DTs are illustrated in Fig. 3. The key research methods in fault diagnosis are as follows:

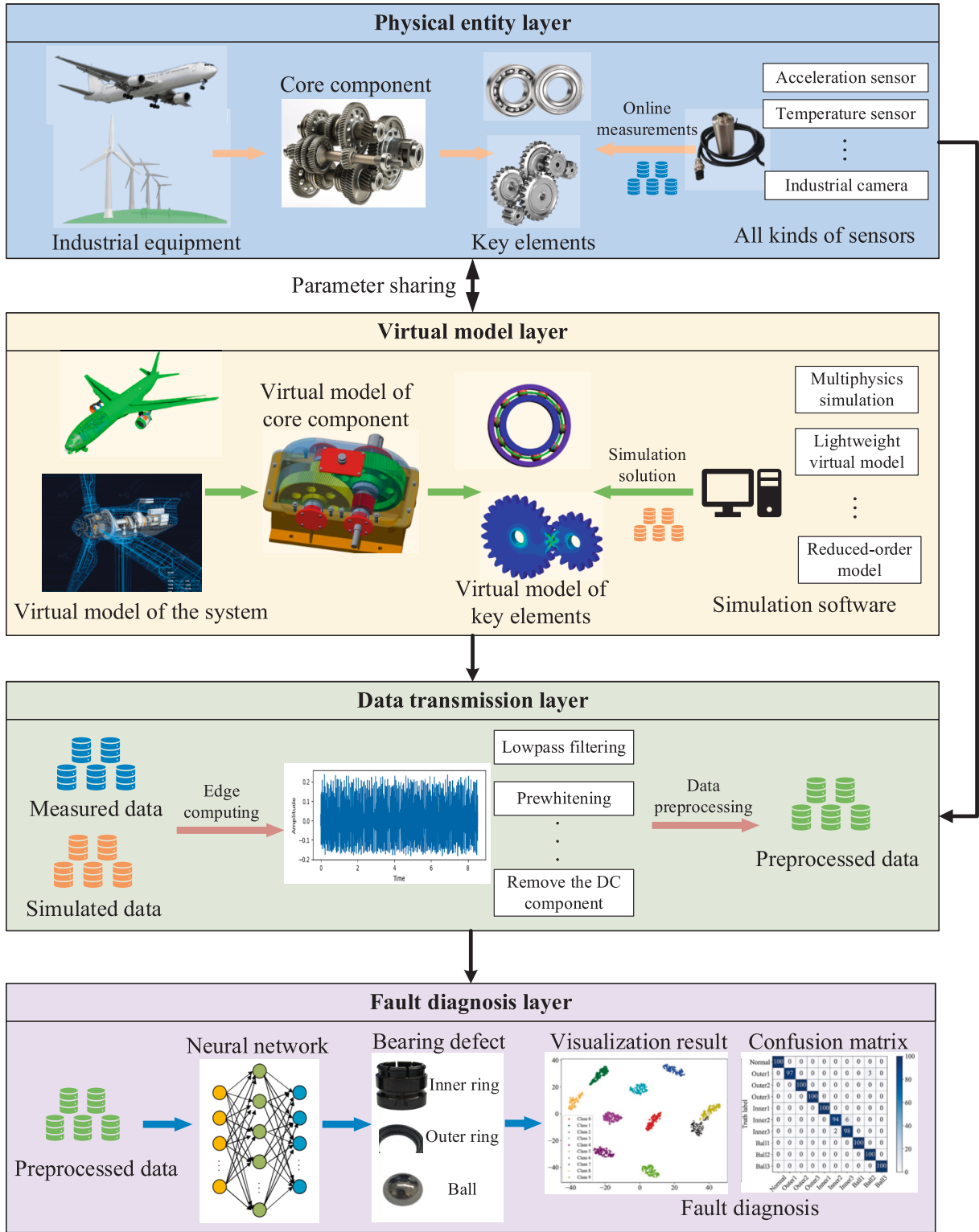


Fig. 1. The architecture diagram of engineering application and deployment of DT-assisted fault diagnosis.

- (1) **High-fidelity:** A defining feature of DTs is the creation of high-fidelity virtual models, which is essential for achieving a true digital twin. Building an accurate virtual model at the initial stage is challenging. Therefore, the virtual model and its parameters need to be continuously optimized and updated to obtain a high-fidelity digital model.
- (2) **Real-time performance:** Compared to traditional data acquisition and transmission methods, DTs require real-time data

transmission and interaction to ensure that the virtual model can dynamically and accurately reflect the condition of the physical entity.

- (3) **Closed-loop control:** The data transmission between the physical entities and their twin models is bidirectional and enables mutual control. The goal of DT is to achieve precise mapping and mutual control between the physical entities and their twin models, and to accurately predict future conditions.

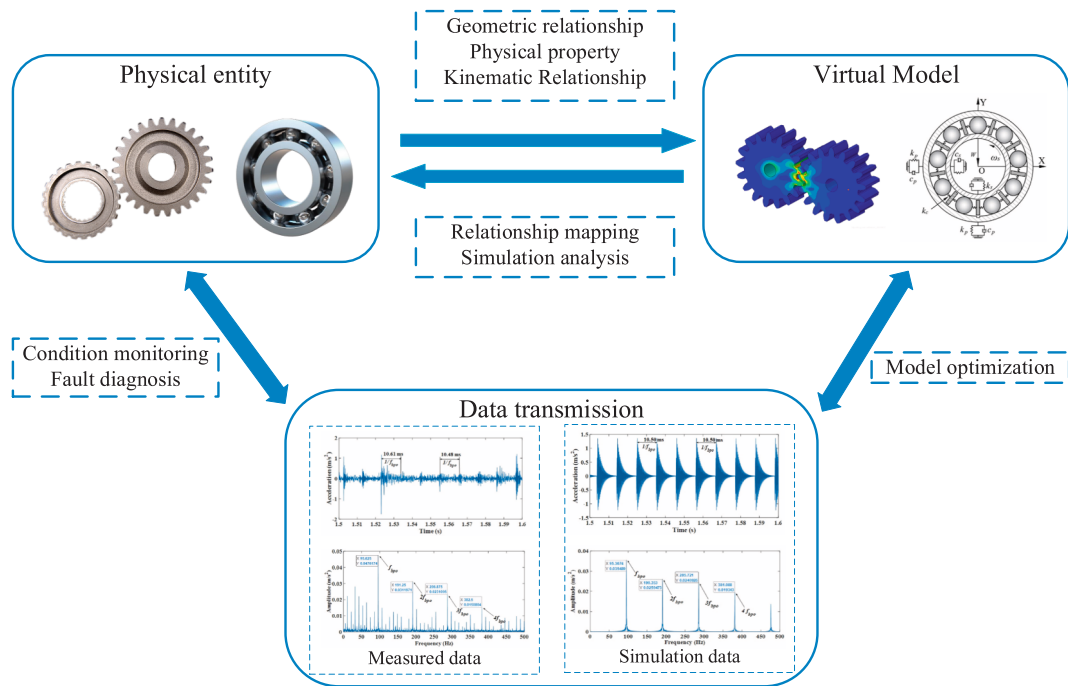


Fig. 2. The general framework for the DT system.

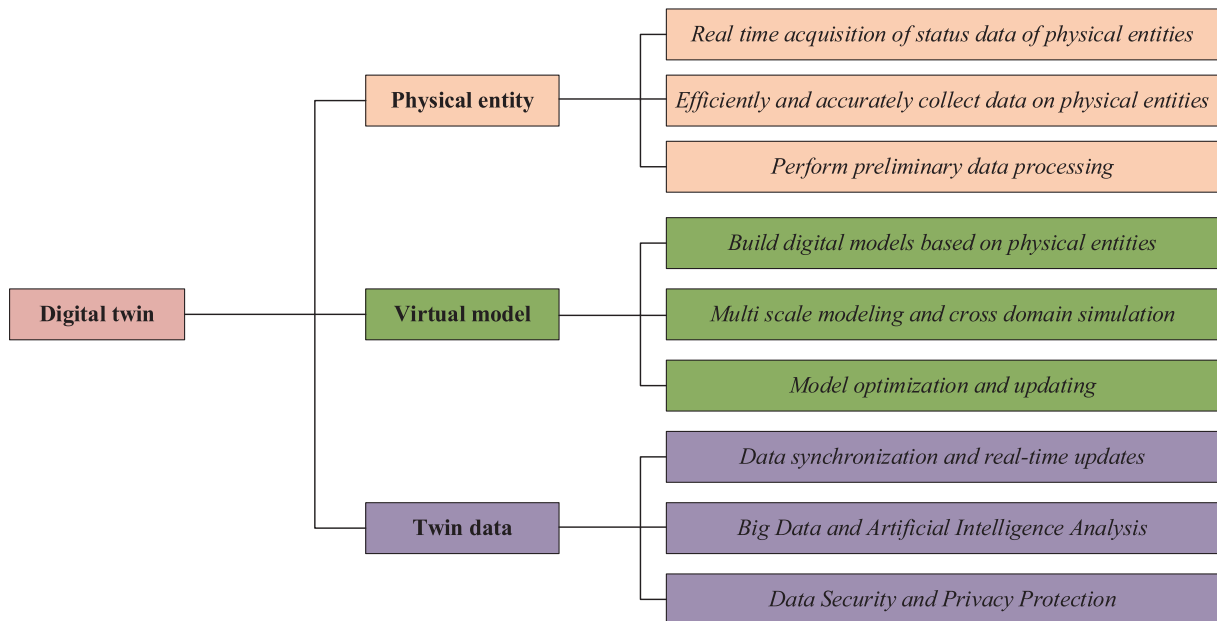


Fig. 3. Key research methods of the DT system.

DT is emerging as a powerful fault diagnosis tool and will focus on enhancing model adaptability, reducing data latency, and improving control accuracy. With the development of artificial intelligence technology, DTs will play an increasingly important role in IFD, fault prediction and preventive maintenance.

1.3. Motivation

DT technology is being explored in mechanical condition monitoring, fault diagnosis, and related fields. Although the technology is still in its early stages and not yet fully mature, it demonstrates unique advantages. Based on a comprehensive review of the literature, DT

technology enables comprehensive mapping of physical entities through the construction of virtual models and facilitates deep interaction between these models and their physical counterparts. Therefore, DT technology can accurately reflect the dynamic operating characteristics of physical entities. It effectively addresses the issues of insufficient and imbalanced fault data caused by harsh operating environment and complex working conditions, thereby enabling equipment diagnosis even under limited fault data. In addition, the accurate virtual models allow for effective monitoring and prediction of real operating conditions, thus supporting real-time condition monitoring and predictive maintenance of equipment.

As shown in Table 1, some researchers have presented concise

Table 1
Analysis and comparison of review papers.

Ref.	Time	Main contents	Object of the overview	Innovation
[34]	2023.1	●The definition, characteristics and application scenarios of DT were clarified. ●The research methods of physical entities, virtual models and twin data were reviewed. ●The popular application fields of DTs have been summarized.	●Food, agriculture, oil and gas industry, supply chain, indoor safety management, among others.	●The differences between DTs and cyber-physical systems were compared. ●The prospect of establishing a general software platform for multi-domain DT models integrating model construction, testing and implementation was proposed.
[41]	2024.3	●Different levels of fusion-based diagnosis methods for faults in hydraulic components were discussed. ●The DT-assisted IFD method was elucidated.	●Hydraulic components	●Through the review of IFD based on information fusion of hydraulic components, valuable guidance was provided for further research in this field.
[42]	2024.6	●The mechanisms and causes of bearing faults were summarized. ●The types of condition monitoring and fault diagnosis techniques were evaluated and described. ●The time–frequency analysis method and the dynamic linear regression method were introduced for the application of the rolling bearing fault diagnosis and condition monitoring.	●Bearings	●In this paper, bearings were used as an example to evaluate and analyze the current status of condition monitoring and fault diagnosis in rotating machinery.
[43]	2024.11	●The research progress on DT technology is analyzed, including the concept of DT, development history, and the key technologies required for its application. ●The related concepts and development history of IFD, along with the application of ML, DL, and other intelligent algorithms, are described. ●The application of DT technology in IFD was analyzed.	●Industrial manufacturing (bearing, gearbox and motor), energy and aerospace	●In this paper, the concept, development history, and related technologies of DT and IFD were described, and the application of DT in IFD was analyzed and discussed.

Table 1 (continued)

Ref.	Time	Main contents	Object of the overview	Innovation
[44]	2025.3	●This paper summarized the development, application and key technologies of DT, such as modeling, model modification, and interaction. ●The application of DT in fault diagnosis of rotating machinery was reviewed, and the future research directions were outlined.	●Rotating machinery (bearing and gear)	●The developmental origins and industrial applications of DTs were introduced, and the key technologies and advantages of DT-driven fault diagnosis were discussed in this paper.
[45]	2025.4	●The modeling methods for the bearing, gear, and rotor systems in rotating machinery have been summarized and presented. ●The method of parameter update in the mathematical model was introduced. ●The fault diagnosis and remaining useful life (RUL) prediction methods based on DT were reviewed.	●Rotating machinery (bearing, gear and rotor systems)	●This paper provides a comprehensive review of research advancements in IFD and RUL prediction modeling methods, including those based on data-driven techniques and artificial intelligence approaches.

overviews of the concept, key technologies, and applications of DTs. However, their focuses vary, and existing classifications and organizational frameworks are not sufficiently comprehensive. Notably, no systematic review currently integrates diverse research methods, construction methods for DT models, twin data types, key enabling technologies, and practical application scenarios of DT-assisted fault diagnosis in rotating machinery.

In view of the limitations of existing research, this work conducts a thorough and structured review of the research progress in this domain, with three distinctive contributions: (1) it systematically classifies and compares the main research methods and key technologies of DTs in rotating machinery fault diagnosis; (2) it analyzes representative application schemes and scenarios, highlighting their practical relevance; and (3) it identifies current challenges and outlines future research directions, offering actionable insights to promote the deeper and more universal integration of DT technology into fault diagnosis across different devices and operational contexts.

1.4. Overview of this paper

DT has been identified as a critical enabling technology for the future and has attracted significant attention in rotating machinery fault diagnosis due to its potential to ensure system safety, enhance equipment reliability, and reduce maintenance costs. The contributions of this comprehensive review to the development and application of DT in rotating machinery fault diagnosis are summarized as follows:

- (1) A comprehensive overview of the main research methods of DT technology in the rotating machinery fault diagnosis is provided.
- (2) The various DT modeling methods and data types used in fault diagnosis are analyzed and summarized, offering a valuable reference for future researchers.
- (3) The specific applications of DT in fault diagnosis are reviewed, with relevant literature demonstrating that DT technology can significantly enhance diagnostic efficiency.
- (4) The current challenges of DT-based fault diagnosis methods are discussed, and suggestions for future research and development are proposed.

The paper is organized as follows: [Section 2](#) reviews and analyzes the relevant literature. [Section 3](#) discusses the main research methods of DT in rotating machinery fault diagnosis. [Sections 4 and 5](#) summarize the modeling methods and data types used in DT for fault diagnosis. [Section 6](#) presents the key application scenarios. [Section 7](#) examines the challenges and outlines future development directions. Finally, [Section 8](#) provides the conclusions. An overview of the paper structure is

illustrated in [Fig. 4](#).

2. Literature review

2.1. Literature collection method

The literature review was conducted using the Web of Science, Scopus, and IEEE Xplore databases, from which a total of 16,780 papers related to DTs were retrieved. Among these, 459 papers specifically

Table 2

Literature Retrieval.

Retrieve keywords	Website search	Number of papers	Time frame
Digital twin	●Web of Science	16,780	2005–2025
Digital twin, Fault diagnosis	●Scopus	459	2012–2025
	●IEEE Xplore		

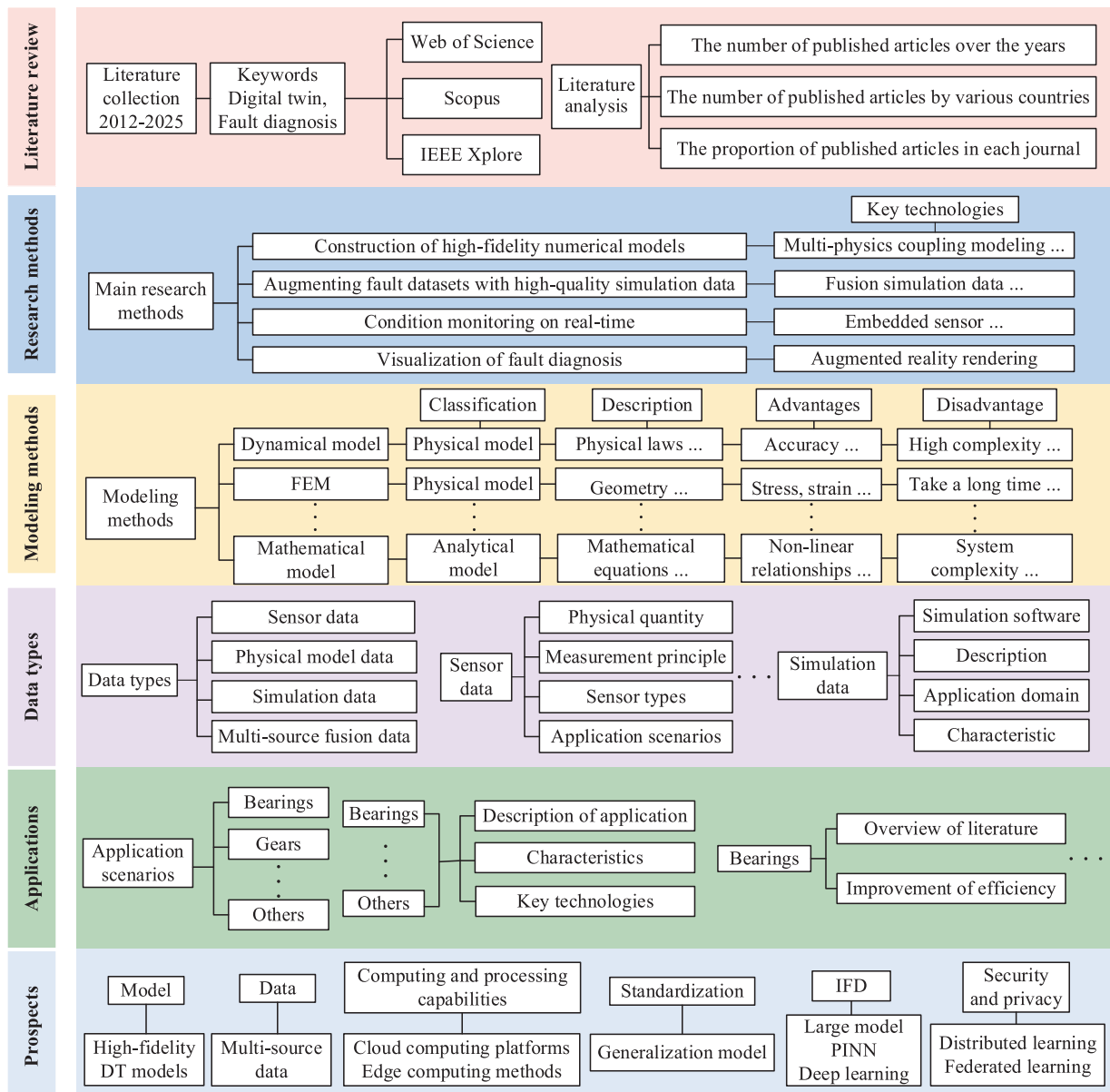


Fig. 4. The overview of the paper.

focus on the application of DTs in fault diagnosis, as shown in Table 2. The concept of “digital equivalents of physical products” was introduced by Michael Grieves in 2003, and the first academic paper on DTs was subsequently published in 2005. Since then, the application of DTs has been explored across a variety of industries, including aerospace, manufacturing, smart cities, product design, logistics and transportation, equipment testing, fault diagnosis, and more. However, the first paper specifically addressing the application of DTs in fault diagnosis was not published until 2012.

2.2. Literature analysis

A statistical analysis of publication trends is conducted in this study, addressing DTs in the context of fault diagnosis, as illustrated in Fig. 5 and Fig. 6. It is observed that the annual number of publications related to DTs remained below 100 until 2017, while those specifically focusing on DTs in fault diagnosis did not exceed 100 until 2023. The continuous increase in the number of publications indicates that DTs have received growing attention in the last two decades. Similarly, DTs in fault diagnosis has emerged as a prominent research topic following this trend.

A total of 238 relevant articles were identified from original 459 publications on DTs for fault diagnosis. To further analyze the research distribution in this field, the countries of origin and publishing houses of publications were systematically classified as shown in Fig. 7 and Fig. 8. The top six countries by number of published articles are China, the UK, South Korea, Norway, the USA, and India. This highlights the emphasis these countries place on advancing DT technology, with China showing particular prominence. Regarding publishing houses, Elsevier has the highest publication volume, accounting for approximately one-third of all DT-related articles in fault diagnosis. This indicates that the publications are high-impact and that Elsevier plays a leading role of DT research.

3. Main research methods of DT in rotating machinery fault diagnosis

As an advanced interaction technology between physical entities and virtual models, DT offers an innovative methodology that integrates data-driven approaches with mechanism-based models for fault diagnosis in rotating machinery. By constructing high-precision dynamic

mapping models, DTs facilitate multi-dimensional analysis and full life-cycle management of rotating machinery, significantly enhancing the real-time performance and reliability of diagnosis. To fully grasp the application approaches and technical paths of DT in this field, this section aims to systematically summarize the main methods of current DT research and analyze their advantages and characteristics across different diagnostic stages. As listed in Table 3, the research methodologies in this field focus on the construction of a high-fidelity digital model, augmentation of fault datasets with high-quality simulation data, real-time condition monitoring, and visualization of fault diagnosis. These approaches overcome the limitations of traditional diagnostic methods in terms of modeling accuracy, data diversity, dynamic responsiveness, and decision-making interaction.

3.1. Construction of high-fidelity digital models

The high-fidelity digital model is the core of DT technology and can accurately describe the geometry, physical characteristics, and dynamic behavior of rotating machinery [46]. Through accurate modeling, a deeper understanding of the operating principles and fault mechanisms of the mechanical system can be achieved, which provides a solid foundation for fault diagnosis. Compared with traditional methods, the high-fidelity digital models offer significant advantages in accuracy, level of detail, and realism. They can more accurately reflect the characteristics of physical entities, thereby providing a reliable basis for fault diagnosis and performance optimization. More importantly, this high-fidelity modeling method provides a new paradigm and technical support for research on intelligent fault diagnosis based on deep learning, facilitating the deep integration of data-driven approaches with physical mechanism models and thereby advancing fault diagnosis technology toward greater precision and enhanced interpretability.

The construction of a high-fidelity dynamic model capable of simulating the real dynamic behavior of a bearing is clearly demonstrated in Fig. 9 through the definition of accurate geometry, material properties, and complex contact mechanics. The process from high-precision geometric import and meshing refinement to the application of material properties and boundary conditions is specifically illustrated in Fig. 10, through which a finite element model (FEM) of the gear is constructed to accurately depict geometric details and physical characteristics. Compared with traditional methods, the high-fidelity digital model

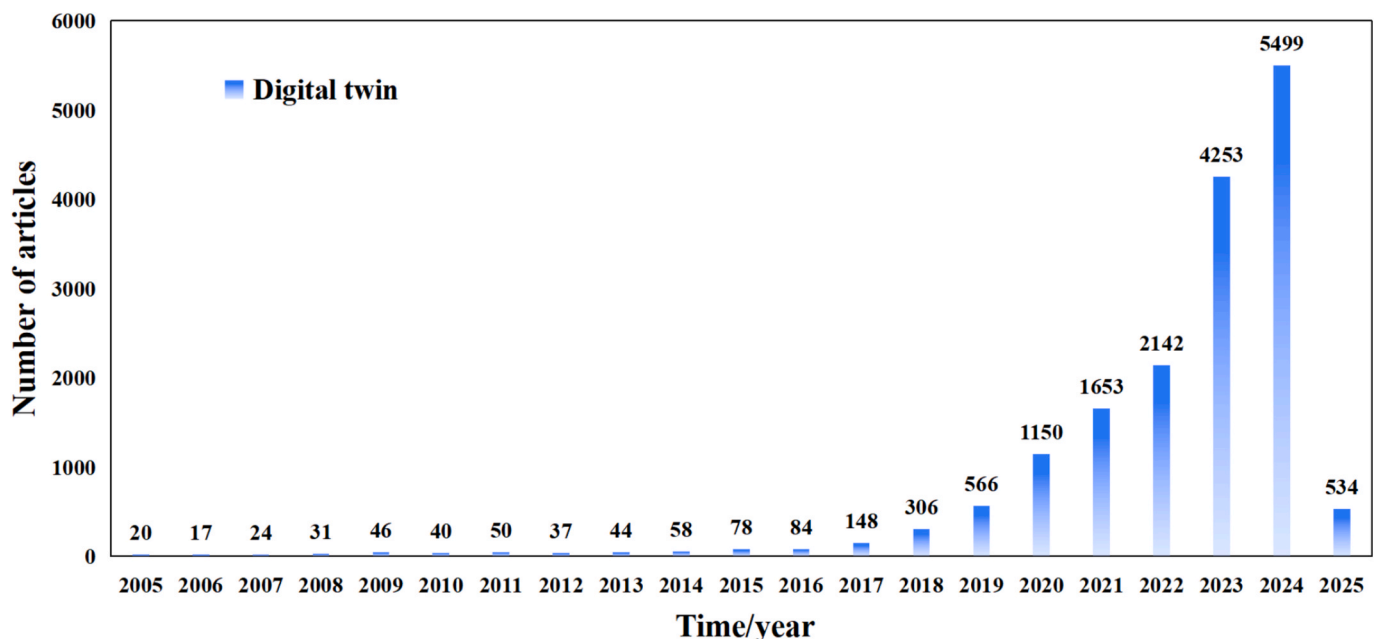


Fig. 5. Trends in the publication of articles related to DTs over the years.

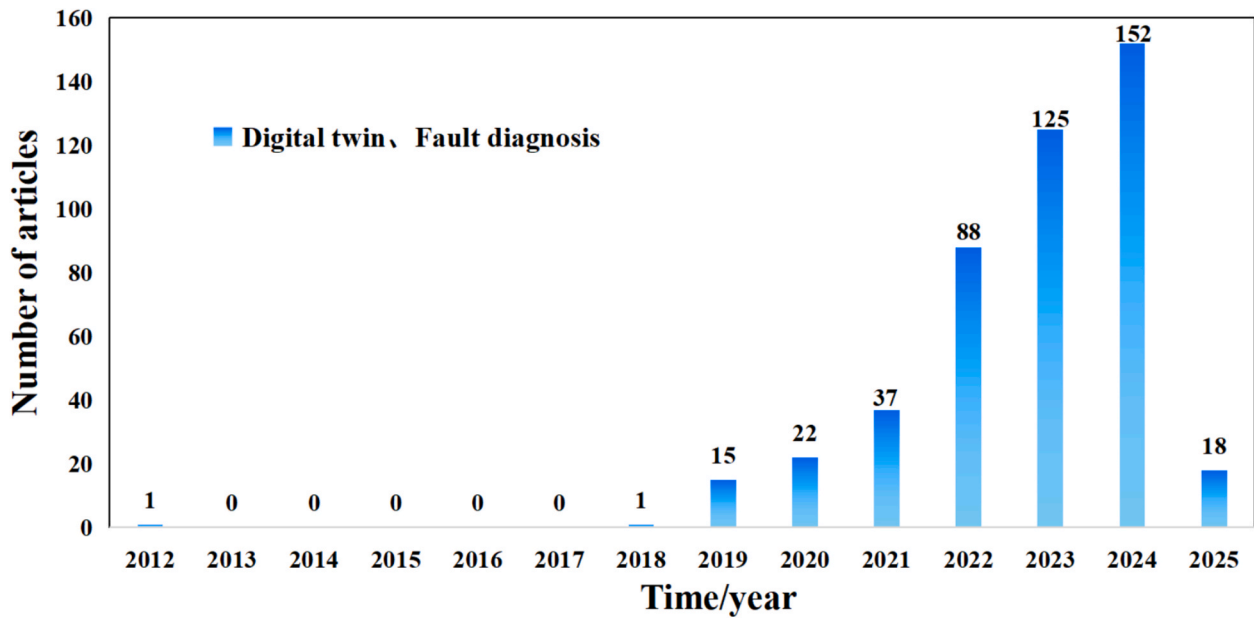


Fig. 6. Trends in the publication of articles related to DTs in fault diagnosis over the years.

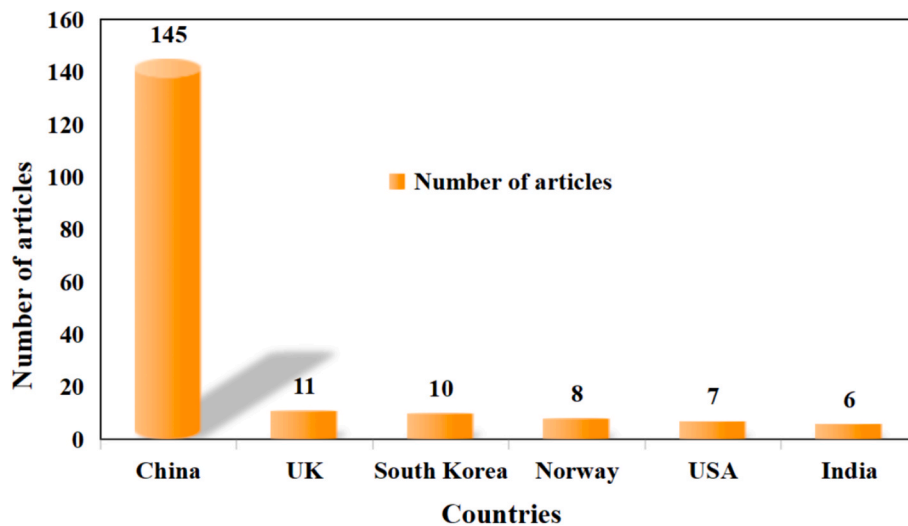


Fig. 7. Number of articles published by each country.

offers significant advantages in accuracy, level of detail, and authenticity. Shi et al. [47] established a nonlinear dynamic model of cylindrical bearings considering the evolution of defects, which provides a strong candidate for DT modeling of bearings in condition monitoring. Song et al. [48] developed a dynamic model of bearings under variable speed conditions, and whose accuracy can be improved by updating speed parameters, stiffness parameters, and fault size parameters. Fan et al. [49] employed the Modelica language to construct a comprehensive DT integration model of rolling bearings across their life cycle. Zhu et al. [50] built a high-fidelity DT model of a two-stage gear transmission system and further updated it with measured normal data. Kang et al. [51] constructed a high-fidelity physical model of the structure of a hybrid drilling robot. Han et al. [52] proposed a high-fidelity DT modeling method of gearboxes based on a knowledge graph coupled with multi-physics. Hu et al. [53] designed a new architecture termed the Ramanujan DT, which is used to construct the high-fidelity simulation model for every time frame throughout the entire life cycle of rotating machinery. Liu et al. [54] proposed a sparse polynomial chaotic expansion method to generate dynamic-polymorphic uncertain DTs,

which maintained a balance between model accuracy and computational complexity. These advancements mark the transformation of high-fidelity modeling from static description to dynamic cognition, providing the foundational basis for the next generation of intelligent diagnostic systems integrated with DTs.

3.2. Augmentation of fault datasets with high-quality simulation data

Data-driven DL and other neural network methods have demonstrated high performance in rotating machinery fault diagnosis over the past decade [55]. Classic models, such as long short-term memory (LSTM) [56–58], convolutional neural networks (CNN) [58–60], denoising autoencoders (DAE) [57,61–63], diffusion models [64–69], generative adversarial networks (GANs) [70–74], deep belief networks and their variants [75–77], have achieved high diagnostic accuracy and wide applications. Some studies have fused multi-sensor features to overcome the limitations of single-sensor data and improve the accuracy of fault diagnosis. However, it is difficult to obtain sufficient fault samples under steady-state operating conditions, resulting in

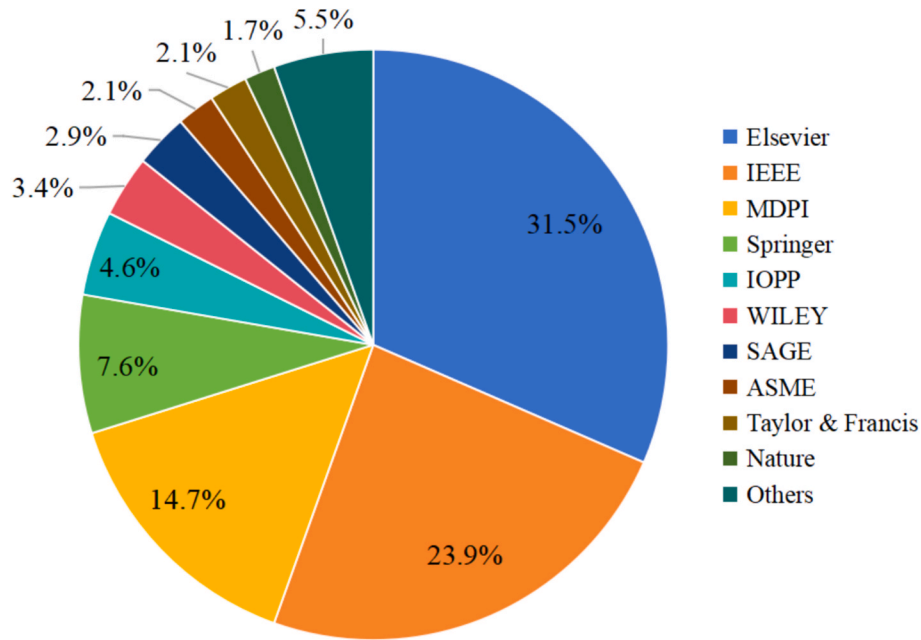


Fig. 8. Publishing houses of articles related to DT in fault diagnosis.

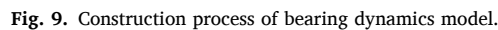
Table 3
Main research methods of DT for fault diagnosis in rotating machinery.

Methods	Key technologies	Limitations	Objectives
Construction of high-fidelity digital models	●Multi-physics coupling modeling ●Adaptive parameter optimization	●The model has high uncertainty. ●Strive to achieve an optimal balance between computational complexity and model accuracy.	●A multi-scale dynamic model established with physical mechanisms is consistent with measured data.
Augmentation of fault datasets with high-quality simulation data	●Multi-condition fault model ●Fusion simulation data	●Both the multi-condition fault model and the fault data are insufficient. ●There are differences between the simulated and the measured data distribution.	●The objective is to generate multi-condition and multi-fault mode labeled data to improve the model's generalization ability.
Real-time condition monitoring	●Edge-cloud collaborative computing ●Lightweight model ●Embedded sensor	●There is latency in data transmission. ●Model transfer under dynamic conditions is challenging.	●Achieve accurate abnormal state detection and residual useful life prediction.
Visualization of fault diagnosis	●Augmented reality rendering ●Virtual reality technology ●visualization technology	●High-fidelity real-time rendering demands substantial computing power. ●Visual interaction with multi-source data fusion has limitation.	●The enhancement of interpretability in diagnosis results and the augmentation of man-machine collaborative decision-making capabilities are pivotal objectives.

unbalanced data, that significantly reduces the performance of most models based on neural network methods such as DL. Moreover, the implementation of these methods is often hindered by the challenge of

acquiring high-quality data in practical engineering scenarios [2]. However, the implementation of these approaches is often hampered by the challenge of acquiring high-quality data in practical engineering. By constructing high-fidelity virtual models for simulation and analysis, DT is regarded as a potential approach to address these limitations. It can help generate realistic simulation data to mitigate the issues caused by limited datasets and imbalanced data distribution. Changes in the parameters of the virtual model can flexibly simulate multiple types of fault data for training neural networks, greatly improving the accuracy of fault diagnosis.

Liu et al. [78] proposed a novel approach to the fault diagnosis of reducer gears by leveraging the high-fidelity behavior modeling and data mining capabilities of DTs and DL. An intelligent diagnosis method for rotating machinery faults was established by Xia et al. [79], which was based on a DT model. It was demonstrated to be capable of generating virtual vibration signals corresponding to multiple health conditions. Based on physical-virtual data fusion, a DT-driven gearbox fault diagnosis method was proposed by Xia et al [80] to address the challenge of insufficient fault data, with the aim of enhancing diagnostic performance through the integration of physical and virtual data sources. Yan et al. [81] established a dynamic simulation model, which assisted the DT in generating high-quality simulated fault data. Azari et al. [82] addressed the problem of the scarcity of training data for IFD labelling by employing DT to generate supplementary datasets. To mitigate the scarcity of fewer fault samples in industrial application, Shang et al. [83] proposed a DT-assisted bearing fault diagnosis method based on virtual-real mapping. Gong et al. [84] proposed a DT-assisted approach to address the issue of limited measured data for bearing fault diagnosis. Shi et al. [85] established a model-data combination-driven DT model (MDCDT) for fault diagnosis of rolling bearings with fewer samples. Sow et al. [86] integrated bearing flexibility into a DT to generate reliable data for ball bearing diagnostics by supervised ML. A bearing failure DT model was developed by Liu et al. [87] to generate fault twin data for test bearings under varying health conditions. These achievements mark the transformation of fault diagnosis from data-dependent scarcity to model-data dual-driven: the virtual data generated by DT can expand small-sample training sets, while the hybrid-driven framework ensures the transferability of the diagnostic model to real-world devices. The general process of enriching a rolling bearing diagnostic dataset using DTs is shown in Fig. 11.



Li et al. [98] developed a data-driven virtual monitoring system using DT technology for real-time turbine rotor operation, enabling virtual control of the rotor operation process from both physical and geometric perspectives. Hu et al. [99] proposed a methodology for fault diagnosis in magnetic levitation bearings by integrating DT with neural networks for achieving continuous real-time monitoring and fault diagnosis. Putranto et al. [100] combined DT with robotics to enable accurate and reliable sensor deployment in complex environments, which provides a scalable solution for structural health monitoring in industrial applications. Purcell et al. [101] constructed a DT model to intelligently monitor the operating status of marine propulsion components based on real-time propeller fluid dynamic torque. Rojas et al. [102] reviewed the application of DT for early fault detection through real-time monitoring. Karyofyllas et al. [103] improved the efficiency

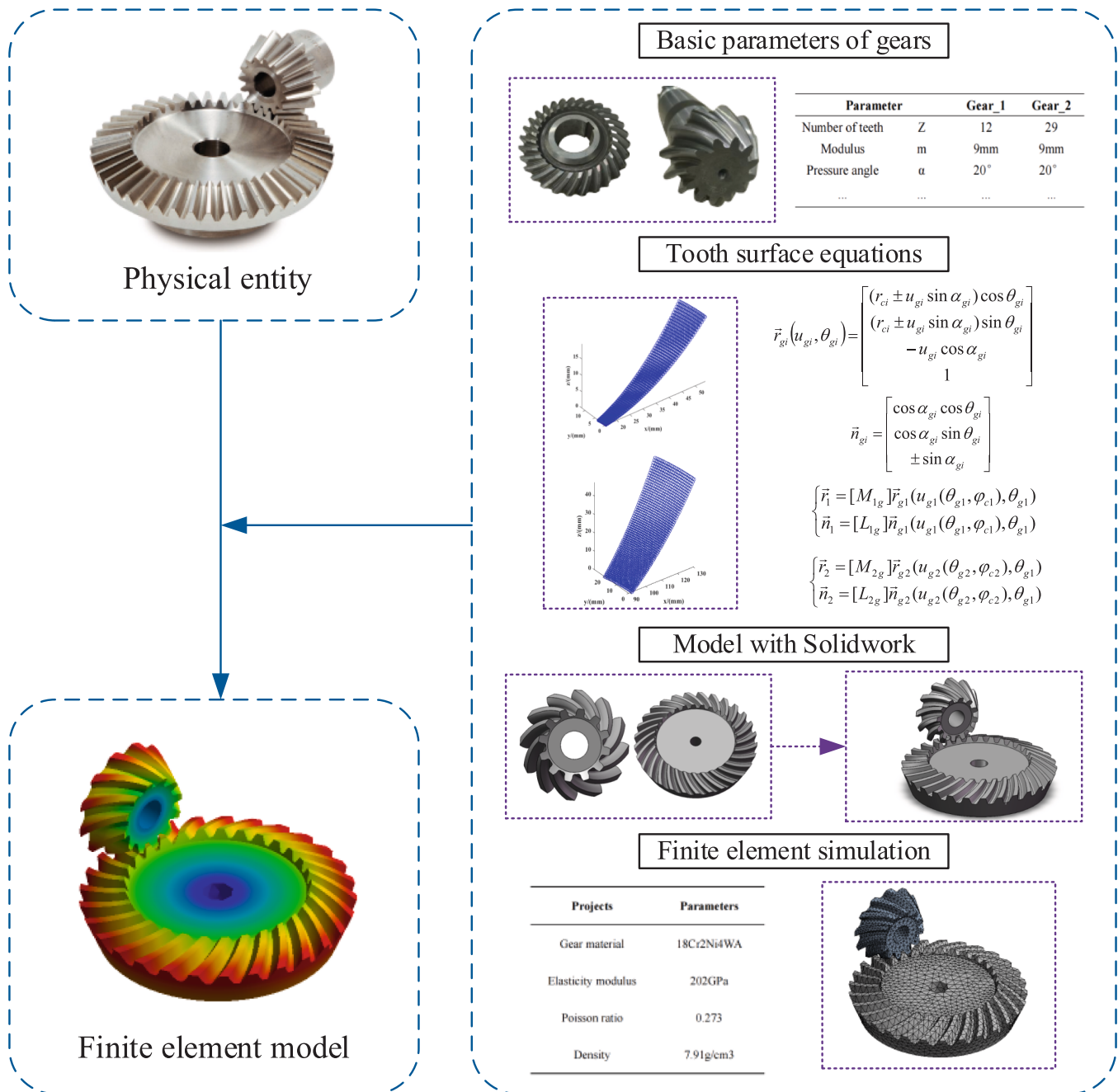


Fig. 10. Construction process of gears models by FEM.

and accuracy of condition monitoring of trusses in the laboratory by adopting data-driven modeling. Bull et al. [104] developed a probabilistic DT model for offshore wind turbines, using a structural health monitoring system during operation to determine the optimal risk-based inspection plan. These advancements indicate that DT real-time monitoring technology has developed adaptability to diverse scenarios. In applications ranging from precision rotors to large-scale wind power, the core lies in the customization of virtual models according to equipment characteristics and the realization of closed-loop monitoring through the collaborative design of algorithms and hardware. The current challenge is how to strike a balance between model complexity and the real-time performance of edge computing. The general process of monitoring equipment operation in real-time with DT is shown in Fig. 12.

3.4. Visualization of fault diagnosis

The DT method is characterized by building a virtual model using real-time data, which accurately replicates the operating state of the physical equipment. By establishing a DT platform, the operating state of the physical equipment can be continuously monitored in real time, and faults can be visually diagnosed through the virtual model. Visualization of fault diagnosis helps engineers quickly understand and analyze equipment faults by presenting complex diagnosis results in an intuitive manner. By using the DT platform, the operating status, fault location, and severity of the equipment can be displayed in the form of three-dimensional graphics and thermal maps to provide a clear view of the fault. This approach not only improves the efficiency of fault diagnosis, but also enhances the understanding of fault causes and provides support for decision-making and maintenance planning. Therefore, this

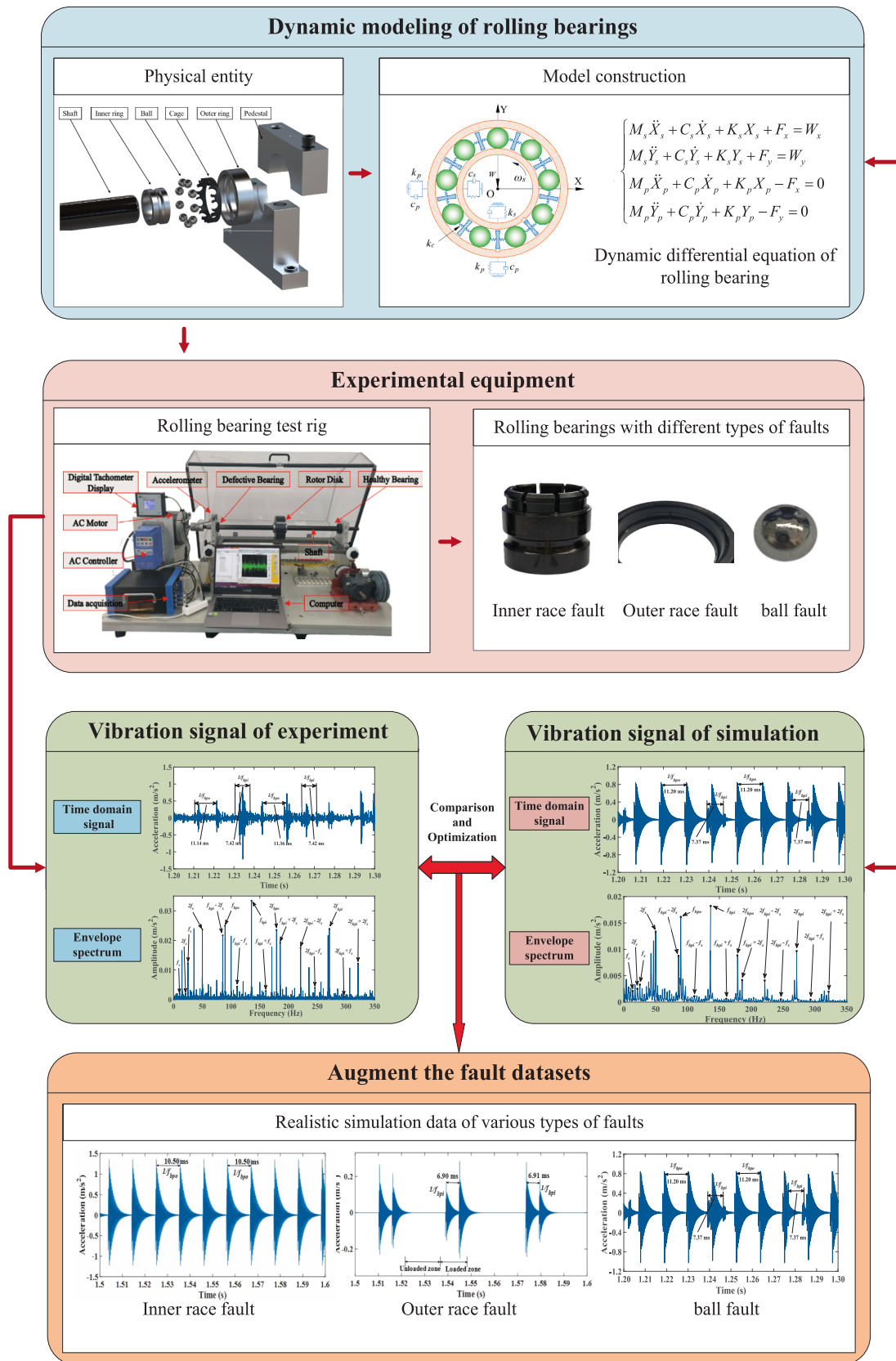


Fig. 11. Process of augmenting fault datasets with DTs: a case study of rolling bearing.

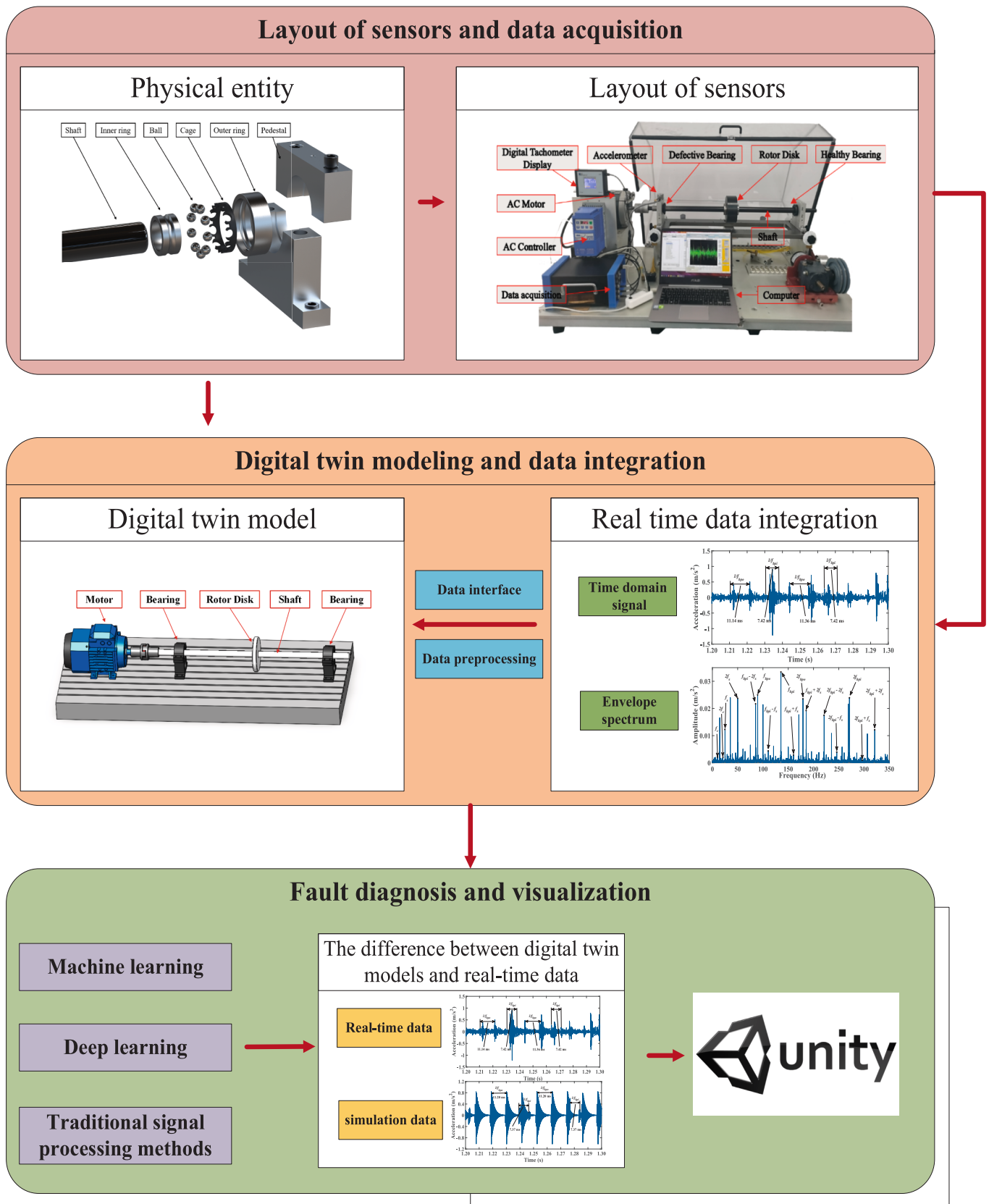


Fig. 12. Process of monitoring equipment operation in real-time with DTs: a case study of rolling bearing.

section summarizes the research progress of digital twin (DT) in visual fault diagnosis. By building a three-dimensional dynamic visualization platform that establishes a mapping between the virtual and the real worlds, a closed-loop transformation from raw data to actionable decision-making knowledge is achieved, providing an intuitive and efficient solution for the intelligent operation and maintenance of industrial equipment.

Ma et al. [105] employed a centrifugal chiller DT system for real-time fault diagnosis using 3D visualization and monitoring. Su et al. [106] proposed a condition monitoring and visualization framework of a marine cooling pump based on a DT model. Liu et al. [107] integrated DT into the wind turbine transmission system fault diagnosis to enable real-time monitoring and simulation of actual operating states. Hasan et al. [108] applied a DT method to collect data from various sensors on board the ship, perform comprehensive analysis, calculate failure parameters, and provide real-time visualization of the results. Yao et al. [109] proposed a transformer fault diagnosis method based on the integration of DT technology, utilizing visualization techniques to represent multidimensional DT models, thereby facilitating IFD and health monitoring. These cases confirm the model-data dual-driven characteristics of DT visual diagnosis, providing a technical paradigm for fault diagnosis that combines both intuitiveness and accuracy. The current bottleneck lies in balancing massive monitoring data with the efficiency of real-time visualization rendering.

In addition, there has been a large amount of research devoted to integrating DT with advanced visualization techniques to achieve real-time monitoring of equipment operation and visual fault diagnosis. This methodology has been demonstrated to provide substantial technical support and guidance for application scenarios such as unmanned digital factories [110–112], complex network systems [113], and subsea production systems [114,115], where fault diagnosis is particularly challenging.

4. Modeling methods of DTs in rotating machinery fault diagnosis

The core of DT lies in the construction of high-fidelity digital models. However, due to technological limitations, the development of these models requires careful consideration of the trade-off between the computational complexity of physical systems and the efficiency of simulation performance. It is imperative that the virtual model is designed to accurately reflect the geometric, material and dynamic properties of the physical entities, while considering the computational constraints imposed by both hardware and simulation software. A research paradigm for virtual models, including geometric, physical,

behavioral, and rule-based models has been proposed by Tao et al. [116]. Although interdisciplinary modeling approaches are diverse, the proposed virtual model framework can effectively reflect the complex properties of physical entities. The application of this framework helps balance the trade-off between physical system complexity and simulation performance efficiency in the process of model development [34]. This section systematically summarizes the core methods used in high-fidelity modeling for digital twin (DT), providing a theoretical framework and practical engineering guidelines for overcoming the typical trade-off between model fidelity and computational efficiency. Several representative DT modeling methods are presented in Table 4.

4.1. Dynamical model

A dynamical model is defined as a specialized mathematical representation describing the temporal evolution of a physical and engineering system, which can effectively capture temporal variations and predict future conditions. Notably, these models are particularly well-suited for the analysis of nonlinear and highly complex phenomena. Consequently, dynamical methods are frequently adopted for the construction of DT models. This section systematically summarizes the relevant research on dynamic models in DT. The unique capabilities of dynamic modeling make it an indispensable approach for addressing transient processes and strongly coupled systems in DT applications.

Shi et al. [85] proposed a DT model based on a combination of model-driven and data-driven approaches, which is predicated on the dynamic behavior of rolling bearings. Similarly, Liu et al. [87] developed a fault diagnosis DT model based on a rolling bearing dynamic model to address the challenges of data collection and labeling in practical engineering. Song et al. [97] constructed a dynamic model of robot joints to generate simulation data, which was employed as source domain data for the DT model. Tai and Altintas [117,118] constructed a dynamic model for the bearings in the spindle system of machine tools to simulate vibration signals caused by bearing faults, worn contact interface between the shaft and tool holder, and spindle unbalance. Wu et al. [119] established a dynamic model of rotating blades for an aero-engine, and proposed a blade collision fault diagnosis method based on ESKESED. Li et al. [120] constructed a four-degree-of-freedom dynamic model of a rolling bearing to simulate the physical behavior of the bearing. Souleymane et al. [121] developed a dynamic model of a gearbox to generate simulation data under various operating conditions. Liu et al. [122] utilized a bearing dynamic model to construct a DT framework and acquire the necessary data for train fault diagnosis. These cases demonstrate that DT technology based on dynamic models offers a reliable solution for intelligent fault diagnosis, overcoming the

Table 4
DT modeling methods in rotating machinery fault diagnosis.

Modeling methods	classification	Description	Advantages	Disadvantages
Dynamical model	Physical model	●It is a model based on physical laws that describes the motion of a mechanical system under the action of forces, moments, or other incentives.	●The model has the ability to accurately reflect the physical characteristics of the system.	●The model exhibits high complexity, demands significant computational resources, and depends on detailed mechanical parameters.
Finite element model (FEM)	Simulation model	●The rotating machinery was discretized by the FEM to simulate the stress, strain, and dynamic behavior.	●It can accurately describe complex geometric structures and material properties, and is widely used in engineering fields.	●It demands a large amount of computation and adequate computing resources.
Simulink model		●Using the Simulink platform, the dynamic simulation is carried out by establishing a mathematical model and simulating the physical process of the system.	●The interface is intuitive, suitable for system-level modeling and simulation, and easy to debug and optimize.	●The accuracy of the model is contingent upon the settings and parameters of the model itself, and it may not be suitable for engineering applications that require high precision.
Mathematical model	Analytical model	●The dynamic behavior model of rotating machinery is constructed based on mathematical equations.	●It enables simple and intuitive rapid analysis and modeling.	●Some complexity of the system may be overlooked, and model accuracy is limited.
Machine learning model	Data-driven model	●Based on a large amount of historical data, the fault diagnosis model of rotating machinery is constructed by ML algorithms.	●It can handle complex non-linear relationships without precise physical modeling.	●It requires a large amount of high-quality data.

bottleneck of scarce fault samples in practical engineering. Integrating the time-varying characteristics of dynamic models with the generalization capabilities of data-driven methods is redefining the technical framework for intelligent operation and maintenance, though reducing the real-time computational overhead of high-dimensional models remains a key challenge. The typical DT framework is shown in Fig. 13.

4.2. Finite element model

The FEM is a numerical method for solving partial differential equations in complex engineering and physical problems. It is widely used in solid mechanics, fluid mechanics, heat transfer, electromagnetics, structural analysis, and materials science. A significant trend involves integrating FEM simulation software, such as ANSYS, Abaqus, COMSOL Multiphysics, and MSC Nastran, into DT. This section systematically summarizes the research progress on the FEM in DT. Compared with other methods, FEM offers significant adaptability to complex geometries, robust multi-physics field coupling capabilities, and a mature industrial software ecosystem, making it the preferred approach for constructing high-fidelity DT models—particularly in engineering scenarios dominated by geometric and material nonlinearities.

Lee et al. [88] developed a FEM model of a permanent magnet synchronous motor using COMSOL Multiphysics to simulate the dynamic response of the motor under healthy and fault conditions. A multi-physics high-fidelity DT model of a permanent magnet synchronous motor is shown in Fig. 14. Yu et al. [123] established a FEM model of a gearbox using virtual prototyping technology and multi-body dynamics software. Ma et al. [124] constructed a FEM model of a bearing test rig for the DT using ANSYS software. Nian et al. [125] developed a FEM model of a gas pressure regulator and utilized Fluent software to perform a fluid dynamics simulation. Feng et al. [126] used the FEM method to

construct a DT model of a bearing and a motor based on thermal characteristics, aiming to achieve causal diagnosis of the remaining useful life (RUL) of high-speed spindle system. Huang et al. [127,128] used FEM to construct a reliable DT model of a rolling bearing under effective loading conditions to generate reliable simulation samples under various operating conditions. Zheng et al. [129] built a reduced-order DT model of a marine centrifugal pump using FEM, and conducted CFD simulations of the model with Fluent. Li [130] established a DT model of a large-diameter auger using FEM and described its physical properties with ANSYS. Collectively, these studies demonstrate that the FEM effectively addresses the trade-off between model accuracy and computational efficiency in DT, striking an optimal balance between physical fidelity and engineering practicality.

4.3. Simulink model

Simulink is an extension of MATLAB that enables users to construct system models by dragging and dropping various predefined functional modules directly. It is used to create DT models, providing a convenient way to facilitate the integration of transportation tools and thus enabling real-time mapping between DT models and physical entities. This section provides a systematic summary of research on Simulink in DT modeling. Owing to its modular and visual development features, Simulink is regarded as the preferred solution for modeling complex electromechanical systems in DT applications.

The DT models of induction motors [89] and triplex piston pumps [131] were constructed using Simscape modules in Simulink. Xie et al. [132] established a DT model of a rolling bearing in Simulink, based on a set of dynamic equations. Fera et al. [133] developed a Simulink model for an induction motor in a marine propulsion system to analyze the behavior of stator windings during a short-circuit event. Bo et al. [134] built a DT model of a diesel engine using MATLAB/Simulink, and

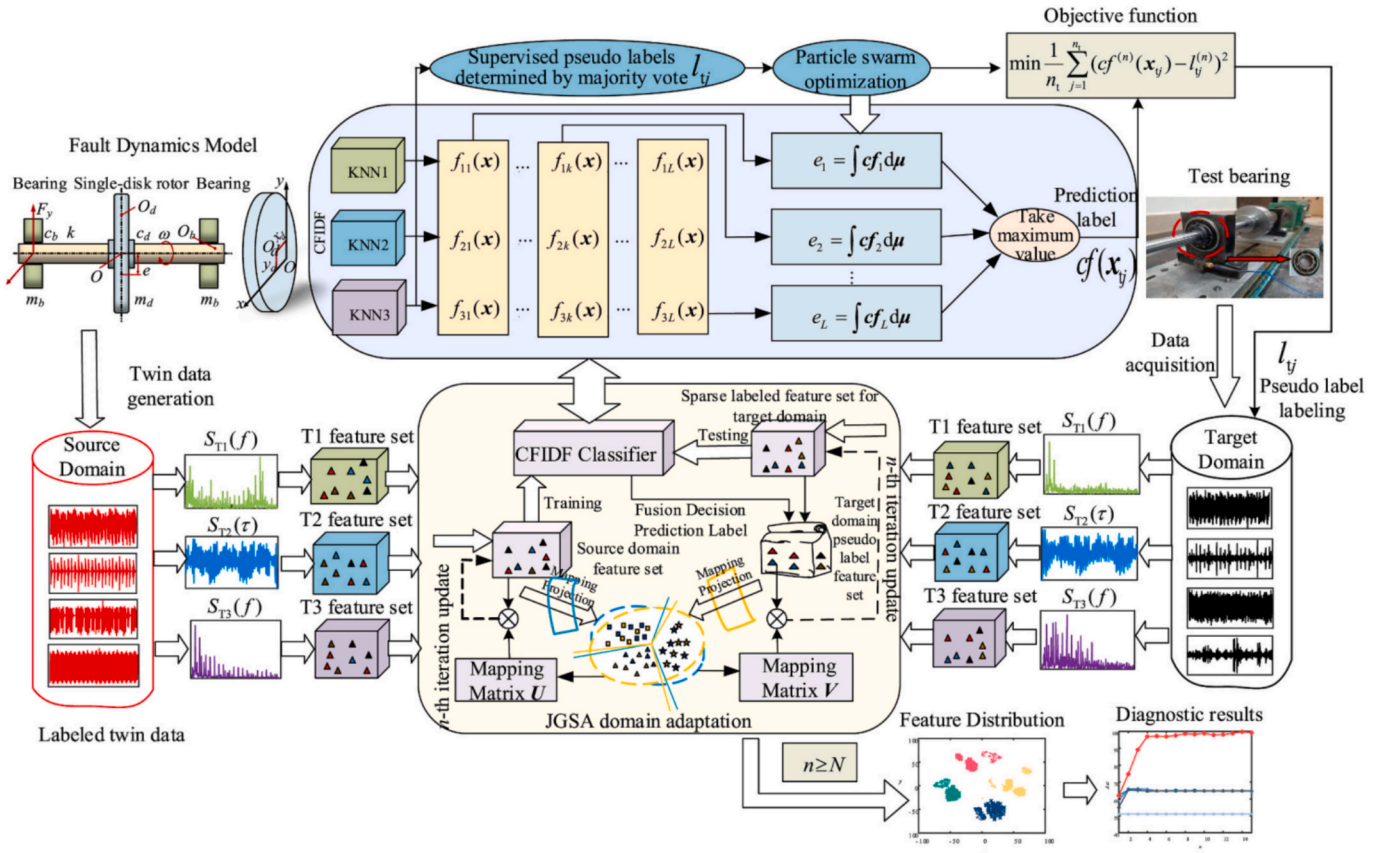


Fig. 13. Framework of DT-based fault diagnosis for rolling bearings [87].

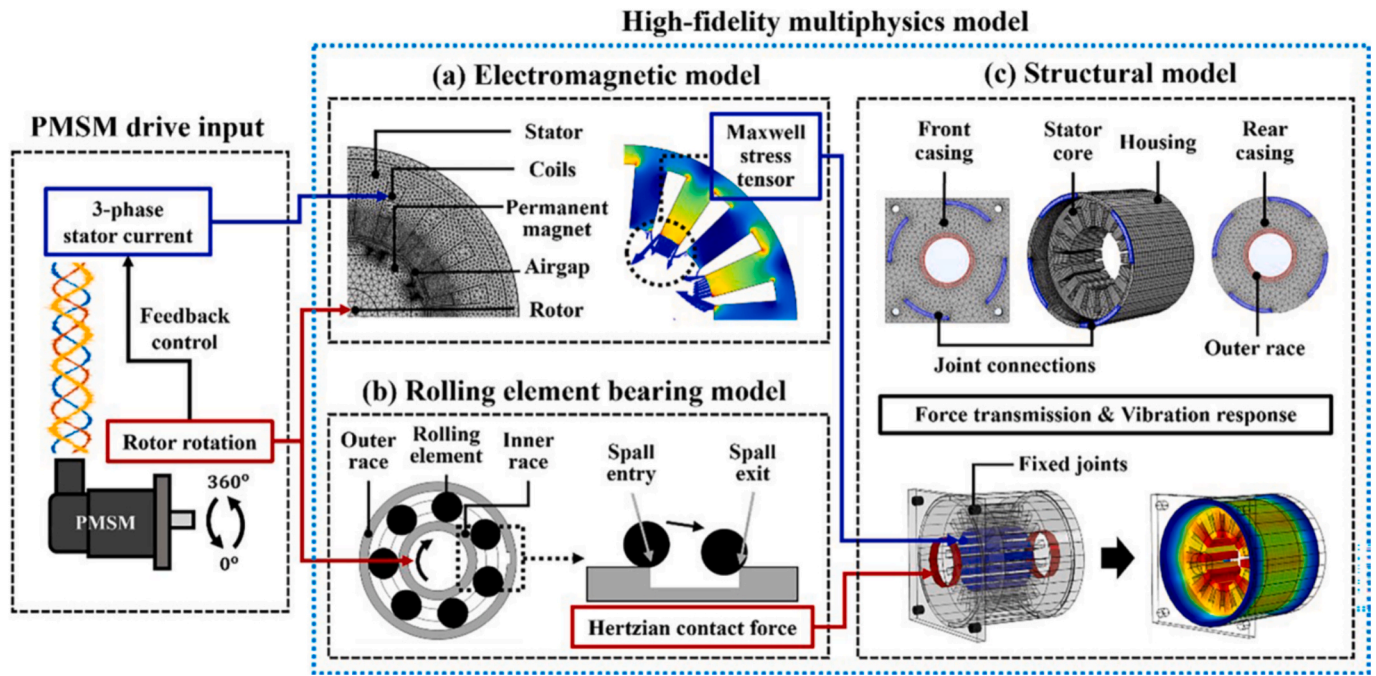


Fig. 14. Flowchart of FEM modeling of a permanent magnet synchronous motor involving multiple physical fields [88].

integrated it with an optimization algorithm to achieve real-time fault diagnosis. Zayed et al. [135] constructed a Simulink model of a three-cylinder pump system using MATLAB and utilized the simulated DT model to study faults such as leakage, blockage, bearing failure, and others. Venkatesan et al. [136] implemented health monitoring and prediction for a permanent magnet synchronous motor by establishing an intelligent DT model in the MATLAB/Simulink environment. Bhatti and Singh [137] developed a DT model of linear actuators for multi-physics robots based on MATLAB/Simulink and integrated it with a custom-designed fault detection mechanism using Naive Bayes classifiers. As shown in Fig. 15, the DT model of an induction motor was constructed using Simulink. The above-mentioned research indicates that the Simulink model significantly reduces the development difficulty of DT for complex equipment by directly converting physical laws into executable modules, and it has been widely applied in DT development.

4.4. Mathematical model

Mathematical models are widely recognized as crucial tools for describing the fundamental characteristics and dynamics of real-world systems. By leveraging these components, physical entities can be effectively translated into DT models, providing an intuitive and systematic representation. This section summarizes the relevant research on mathematical models in DT. Compared with data-driven methods, they offer strong interpretability grounded in physical principles. This feature makes them a fundamental approach for DT in scenarios requiring high reliability or operation under extreme working conditions.

Adamou and Alaoui [138] utilized a digital shadow system based on an efficiency model framework to detect and diagnose faults in induction motors. Khawale et al. [139] developed a computationally efficient, high-fidelity hybrid engine model using mathematical equations to simulate actual engine behavior. Peng and Chen [140] proposed a hybrid observer/Kalman filter identification method for fault diagnosis of Unmanned Aerial Vehicle (UAV) rotor engines, which were modeled by an autoregressive model with exogenous inputs. Li et al. [141] constructed a DT system for a bearing by combining a vibration response model and reinforcement learning technique using both the fault pulse and attenuation oscillation functions. Wang et al. [142] combined a

mathematical mechanism model with a data-driven model to propose a new DT framework for an aero-engine, aiming to enable the digitalization of physical systems. Dong et al. [143] established a DT model of a hypersonic flight vehicle through mathematical simulation and used the constructed fault tree model to simulate multiple fault conditions. The above research indicates that digital twin technology based on mathematical models not only retains the causal interpretability inherent to mathematical models but also incorporates data-driven environmental adaptability, providing a new generation of solutions for the intelligent operation and maintenance of equipment with both robustness and generalization.

4.5. Machine learning model

ML models are specialized mathematical models that make predictions, classifications, or decisions based on patterns learned from training data. They can be used to analyze and describe the dynamic relationships within the data, enabling the DT not only to reflect the real-time condition of the physical entity, but also to make more accurate predictions and decisions. It has the capacity to detect potential anomalies or risks in real time, providing early warnings and facilitating corrective actions to enhance system reliability and safety. Dinh et al. [144] established a DT model of a 2 MW doubly-fed wind power generation system based on ML. Hu et al. [63] constructed a masked one-dimensional convolutional autoencoder model for bearing fault diagnosis based on DT-enabled IIoT by using a newly designed AI approach. Kosova and Unver [145] proposed a DTs-based health monitoring system that employs ML methods to diagnose system failures during the early design stage. Rachmawati et al. [146] proposed a sensor-driven fault diagnosis method that integrates ML techniques to investigate temperature imbalances between extruders and printed surfaces. These studies reveal that the fusion of ML and DT technologies plays a significant role in the intelligent operation and maintenance of industrial equipment.

5. Types of twin data for DTs in rotating machinery fault diagnosis

The main research on DT in rotating machinery fault diagnosis not

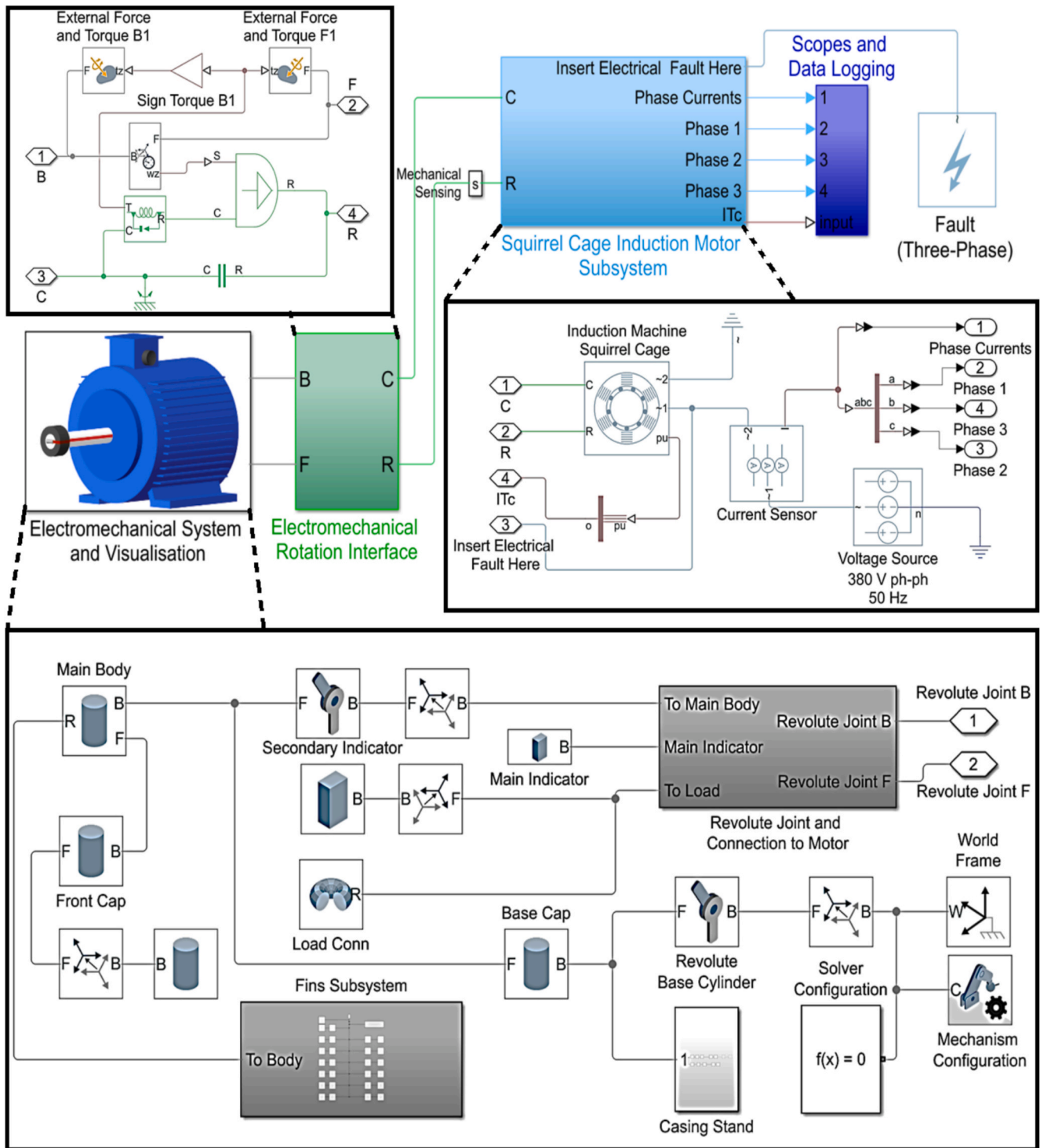


Fig. 15. DT model of an induction motor in Simulink [89].

only focuses on the development of high-fidelity digital models but also encompasses key processes such as data acquisition, processing, and utilization. Twin data provide an accurate and real-time reflection of the physical condition of the system, thereby enabling early fault detection and the development of effective maintenance strategies. This capability plays a crucial role in fault diagnosis. As listed in Table 5, this section systematically summarizes the core data types and technical features of DT in the fault diagnosis of rotating machinery, focusing on the generation mechanisms and value dimensions of four types of twin data: sensor data, simulation data, physical model data, and multi-source

fusion data. The findings provide a standardized data architecture reference for constructing a “data-model” dual-driven IFD system.

The accuracy of DT models is directly influenced by the quality of the twin data, which helps the model better reflect the actual behavior of the system, thereby enhancing the reliability and effectiveness of fault diagnosis. In complex systems, the variety and depth of twin data are critical for capturing subtle behavioral changes, which help to identify abnormal patterns and diagnose potential failures. With the continuous improvement of data acquisition and processing methods, DT models can more accurately reproduce the real-time operating state of the

Table 5

Classification of DT data in rotating machinery fault diagnosis.

Twin Data	Data type	Function
Sensor data	● Classification by measured physical quantity: acceleration, temperature, pressure, current, voltage, etc. ● Classification by output data type: analog signal, digital signal, binary signal.	● Real-time monitoring of the operational status and parameter variations of physical devices and systems.
Physical model data	● Classification by data source: geometric model data, structural mechanics data, thermodynamic data, dynamic model data.	● Representation of the actual physical behavior of devices and systems to facilitate the development of more accurate DT models.
Simulation data	● Classification by physical domain: displacement, stress, velocity, kinetic energy, temperature, etc.	● Simulation of equipment and system operation under various operating conditions to validate and optimize DT models.
Multi-source fusion data	● Comprehensive data generated through multi-source data fusion.	● Enhancing the accessibility and interpretability of data is crucial for developing more comprehensive and effective DT models.

system, thereby improving the efficiency and accuracy of fault diagnosis.

5.1. Sensor data

In DT-based fault diagnosis, sensor data plays a critical role. DTs are fundamentally reliant on real-time, accurate sensor data to enable the continuous monitoring of system states, virtual model generation, and real-time mapping with physical systems. As a critical input to DT models, sensor data is utilized to enhance their predictive and diagnostic capabilities for fault identification. Comparing sensor data with virtual models facilitates the prompt identification of abnormal system behavior, thus enabling the early detection of faults and the implementation of effective maintenance strategies. According to the classification of measured physical quantities, the commonly used sensor data types are shown in Table 6.

As previously discussed, several commonly used sensor types are well recognized. The selection of sensors is determined by the application environment and operating conditions. However, it is generally accepted that the primary function of sensors is to collect real-time operational data and parameters from physical devices and systems. This process is essential for establishing the foundational framework DTs and for effective troubleshooting.

5.2. Physical model data

In DT, the utilization of real physical model data is considered paramount. This data reflects the fundamental physical properties and behavior of the system, thereby providing a reference framework aligned with its real-world counterpart. As a result, the DT model is enabled to simulate and predict the system state with enhanced precision. The integration of real-world physical model data with sensor data is instrumental in the generation of more accurate virtual models, thereby improving the efficacy of troubleshooting and prediction processes. As listed in Table 7, the commonly used types of physical model data include the following.

In addition to the three data types, physical model data can be further categorized into fluid dynamics model data, electrical model data, and other specialized types. These data are derived through the application of precise mathematical formulas and algorithms, which establish the theoretical foundation for DTs. This foundation supports real-time monitoring of system state changes and prediction of potential failures.

Table 6

Sensor data types classified by measured physical quantities.

Physical quantity	Measurement principle	Sensor types	Application scenarios
Temperature	● Thermoelectric effect ● Infrared radiation detection ● Resistance change	● Thermocouples ● Thermistors ● Infrared thermometers ● Thermopiles	● Industrial heating ● Environmental monitoring ● Equipment heat dissipation
Pressure	● Principle of Strain gauge ● Capacitance variation ● Piezoelectric effect	● Piezoresistive sensors ● Capacitive pressure sensors ● Piezoelectric pressure gauges ● Optical fiber pressure sensors	● Force analysis ● Pressure monitoring ● Hydraulic system
Vibration	● Inductance variation ● Piezoelectric effect ● Capacitance variation	● Accelerometer ● Velocity sensor ● Displacement sensor ● MEMS gyroscope	● Mechanical fault diagnosis ● Structural health monitoring
Current	● Hall effect ● Ohm law ● Electromagnetic induction	● Shunt current sensor ● Hall current sensor ● Photocurrent sensor ● Magnetic resistance sensor	● Motor monitoring ● Electronic equipment fault diagnosis
Voltage	● Hall effect ● Electric potential measurement ● Capacitance variation	● Voltage divider ● Hall voltage sensor ● Photoelectric isolation sensor	● Motor monitoring ● Electronic equipment fault diagnosis
Flow rate	● Velocity measurement ● Electromagnetic induction ● Heat conduction ● Ultrasonic time difference	● Turbine flowmeters ● Electromagnetic flowmeters ● Coriolis flowmeters ● Thermal flowmeters	● Flow monitoring ● Hydraulic system fault diagnosis ● Process control

5.3. Simulation data

In DT, simulation data refers to virtual data generated through computational models and simulation techniques to replicate and predict system behavior under various conditions. Based on physical principles, mathematical models, and engineering expertise, simulation data is used to simulate dynamic responses, performance characteristics, and potential failure modes. It offers predictions for different operating conditions, facilitating the evaluation of extreme scenarios that are difficult to replicate in real-world operations. The primary advantage of simulation data is its ability to model complex scenarios, enabling the analysis of potential issues during both the design and operational phases. It provides valuable insights into system behavior without the need for physical trials, thereby reducing testing costs and minimizing the risk of system failures. In DTs, simulation data is integrated with sensor data, historical data, and other sources to enhance fault diagnosis, predictive maintenance, and optimization decisions. Several commonly used simulation software tools are briefly introduced in Table 8.

5.4. Multi-source data fusion

The effective implementation of multi-source fused data is crucial for data processing. The goal is to integrate data from different sources to provide a more comprehensive and accurate representation of the system state. This data fusion compensates for the limitations of individual

Table 7
Common physical model data.

Classification	Description	Characteristic	Software tools	Application scenarios
Geometric model data	● It is used to describe the shape, size and topological structure of an object.	● Descriptive features ● Data accuracy ● Data diversity	● SolidWorks ● CATIA ● ANSYS ● SpaceClaim	● CAD, CAE ● Model construction
Structural mechanics data	● It is used to describe the responses of structures subjected to forces, such as stress, strain, displacement, and modal shapes.	● Highly dependent on geometry parameter ● Highly dependent on materials ● Highly dependent on computational simulation ● High spatial coupling ● Time dependence ● Multi-dimensional coupling ● Nonlinearity and complexity ● Large data volume ● Real-time requirements	● Abaqus ● ANSYS Mechanical ● MSC Nastran	● Structural safety assessment ● Mechanical structure design and optimization ● Simulation of material
Dynamic model data	● It is used to describe the response of the system over time, and the data types include displacement, velocity, acceleration, and force.	● Time dependence ● Multi-dimensional coupling ● Nonlinearity and complexity ● Large data volume ● Real-time requirements	● Adams ● Simulink ● LS-DYNA	● Motion control ● Dynamics ● Vibration and noise analysis
Thermodynamic data	● It is used to describe thermally related physical quantities such as temperature distribution, heat flux density, and thermal stress.	● Highly nonlinear ● Dependent state variable ● Cross-scale applicability	● Fluent ● COMSOL ● Icepak	● Simulation of chemical engineering processes ● Battery reaction thermodynamics ● Combustion and heat engine efficiency

data sources and significantly enhances the adaptability and accuracy of the DT model in dynamic environments. Multimodal fusion data typically includes sensor data, physical model data, simulation data, and historical operation data, each of which has unique characteristics and advantages. Through effective integration, the advantages of these different data types can be utilized, thereby minimizing bias that may arise from reliance on a single data source.

The application of multi-source fusion data significantly enhances the capabilities of DTs in fault diagnosis and prediction. Through real-time monitoring and comprehensive analysis, it enables the timely detection of anomalies, improves the accuracy of fault prediction, optimizes maintenance schedules, and reduces operational risks.

6. Applications of DTs in fault diagnosis

Rotating machinery converts kinetic energy into mechanical energy by rotating the rotor, shaft, and other components, thereby driving various processes or enabling power transmission. The main equipment includes motors [147–149], turbines [144,150,151], and pumps [91,152,153], among which rolling bearings [82–85] and gears [80,81,121] are the most critical rotating components. Fault diagnosis and predictive maintenance of rotating machinery are crucial for ensuring the smooth operation of equipment. With the development of technology, DT is playing an increasingly important role in rotating machinery fault diagnosis. In this section, the application scenarios and technical features of DT in rotating machinery fault diagnosis are summarized, including bearings, gears, motors, pumps, and turbines. These examples provide valuable references and application paradigms for relevant researchers, facilitating their mastery of the specific applications and deployments of digital twin technology in their respective industries. The specific application scenarios of DTs in rotating machinery fault diagnosis are listed in Table 9.

6.1. Bearing

As a critical joint in machinery, rolling bearing performance is the key to ensuring the stable and reliable operation of rotating machinery. It is widely used in aerospace, rail transportation, petrochemical industry, and other fields [154–157]. In recent years, DTs have become a significant tool for bearing fault diagnosis.

Li et al. [158] proposed a DT-assisted fault diagnosis method, which incorporated a bearing dynamic model into an improved TL network to

enhance diagnostic performance. Xu et al. [159] introduced a DT-based discriminant graph learning network for bearing fault diagnosis from unlabeled industrial data. Qin et al. [160] developed a DT model of multi-degree-of-freedom bearing faults for generating fault data, which demonstrated excellent performance in fault diagnosis with unbalanced samples. To address the problem of significant discrepancy in information distribution between simulated dynamic response and real data, Li et al. [161] proposed a DT-assisted double transfer rolling bearing fault diagnosis method with information processing and an adaptive model. Li et al. [162] discussed the key technologies of DT in manufacturing and fault diagnosis of aero-engine main shaft bearings, and analyzed the research progress of DT-based approaches for main shaft bearings from manufacturing to life analysis. The diagnostic accuracy of various bearing fault diagnosis methods, as reported in different studies, is compared in Fig. 16. In the figure, the blue bar represents the accuracy of the conventional method, and the green bar represents the accuracy of the DT-assisted method. As shown in Fig. 16, the DT-assisted method shows significantly higher diagnostic accuracy in all the comparison cases, reaching peak accuracies of 94.79 %, 91.81 %, and 83.00 %, which is significantly better than the performance level of the traditional method. Integrating DL and TL networks, DT technology for bearing fault diagnosis effectively overcomes challenges such as insufficient data labeling, sample imbalance, and distributional discrepancies between synthetic and real data, thereby enhancing diagnostic accuracy. Future research can focus on refining the dynamic response simulation capabilities of DT models according to bearing characteristics under diverse operating scenarios, while improving its adaptability to real-world conditions, to provide robust technical assurance for accurate fault diagnosis and reliable operation and maintenance across the entire life cycle of bearings.

6.2. Gear

Gear transmission systems have been widely used in mechanical equipment for their advantages such as compact structure, high efficiency and constant transmission ratio [163]. However, it is worth noting that mechanical equipment faults, especially faults in gear transmission systems, often lead to serious safety accidents. These faults can result from factors such as design defects, manufacturing quality issues, improper operation, or lack of effective maintenance. Therefore, it is essential to investigate the fault mechanisms of gear transmission systems and take effective preventive and maintenance measures to

Table 8
Commonly used simulation software.

Simulation software	Description	Application domain	Characteristic
ANSYS	It is a widely used computer-based engineering simulation tool, which can conduct structural, fluid, electromagnetic and thermodynamic simulations.	- Structural mechanics - Fluid dynamics - Electromagnetic - Heat and energy	- Multiple physical field coupling - High-precision solution - Open source extension - User friendliness
COMSOL Multiphysics	It is a multi-physics field coupling simulation platform based on the FEM, which is particularly suitable for scientific research and high-end engineering scenarios that require interdisciplinary collaborative simulation.	- Chemical Engineering and Materials - Electrical system - Heat and energy - Acoustics and Vibration - Biomedical science	- Multiple physical field coupling - Flexibly adapt to complex problems - Modular toolkit - Real-time interactive simulation
Abaqus	It is an advanced nonlinear finite element analysis software that excels in addressing complex deformation, contact and failure issues under extreme working conditions.	- Structural mechanics - Electrical system - Biomedical science - Heat and energy - Fluid dynamics	- Multi-scale modeling - Multiple physical field coupling - Strong nonlinear analysis ability - Explicit and implicit solvers work together - Powerful user-defined functions
MATLAB/Simulink	Simulink is an integrated graphical system modeling and simulation tool, and it is an comprehensive platform for algorithm development and system simulation.	- Mechanical system - Electrical system - Hydraulic system - Neural network training - Mathematical modeling	- Multiple physical field coupling - Dynamic system modeling - Highly integrated - Rich tool modules - Interdisciplinary compatibility - User friendliness
MSC Nastran	It is a structural simulation software featuring high-precision finite element analysis, renowned for its accurate solutions and military-grade reliability.	- Structural mechanics - Highly nonlinear system - Thermal stress analysis	- Multidisciplinary coupling - High-precision solution - Provide user-defined interfaces

ensure the safe operation [164,165]. DT is increasingly recognized as a key tool for gear fault diagnosis, providing advanced capabilities for monitoring, analysis, and predictive maintenance.

Sow et al. [121] proposed a fault diagnosis method based on the DT, using a digital and experimental gearbox database classified by a Multiclass Support Vector Machine. Yu et al. [123] built a signal fusion model based on DT, which has the capacity to identify minor faults in a planetary gearbox by fusing real and virtual signals. Zhu et al. [166] established a DT multi-body dynamic model of a planetary gearbox to detect compound faults, such as tooth loss of sun gear and tooth loss or pitting of planetary gear, under different working conditions. A DT model was proposed by Hu et al. [167], which was centered around a Wasserstein generative adversarial network. This model was designed to generate high-fidelity virtual samples based on healthy physical samples

Table 9
Application scenarios of DT in rotating machinery fault diagnosis.

Application scenarios	Description	Characteristics	Key technologies
Bearings	DT is used to establish the operating model of bearings, monitor parameters such as vibration, temperature, and speed, and predict performance degradation, fault types, and locations.	It can accurately diagnose early bearing failures, such as cracks, wear, rolling body damage, and other related faults. Vibration analysis and data-driven methods are used for fault detection and location.	- Vibration or sound signal acquisition - Temperature monitoring - Signal processing and feature extraction - Data-driven approaches
Gears	The dynamic model of gears is built based on DT, and the faults such as wear, cracks, and meshing abnormality are identified by monitoring the operating conditions.	It is suitable for identifying tooth surface damage, noise issues, poor meshing quality and other faults in the gear transmission system. Vibration signal analysis, oil debris analysis among other technologies are used for fault diagnosis.	- Vibration analysis - Noise monitoring - Modal analysis - Tooth surface contact analysis - Oil debris analysis - DL and predictive modeling
Motors	DT is used to monitor the speed, load, temperature and current of the motor in real time, and to diagnose electrical and mechanical faults, such as broken rotor bars, stator winding faults, armature faults, bearing faults, and air-gap eccentricity related faults.	The key operating parameters are used to monitor motor condition and diagnose faults by using vibration, temperature and current signal analysis algorithm.	- Vibration monitoring - Temperature monitoring - Current analysis - Dynamic modeling and simulation - ML and adaptive filtering technology
Pumps	DT is used to monitor the pump flow, pressure, vibration, and other parameters, and diagnose the internal wear faults, cavitation, blockage and other faults.	The performance degradation and faults of the pump can be diagnosed by changes in flow, pressure, and other parameters. DTs help optimize pump operating parameters, predict maintenance plans.	- Vibration signal monitoring - Flow and pressure monitoring - CFD simulation
Turbines	DT is used to monitor turbine temperature, pressure, vibration, speed, and other parameters, and diagnose turbine blade damage, thermal effects, and other faults.	It is suitable for early warning of the thermal and mechanical faults, and to avoid major turbine faults with real-time data monitoring and model prediction.	- Temperature and pressure monitoring - Vibration signal analysis - Numerical simulation (CFD, FEM) - Anomaly detection and pattern recognition
Others	DT is employed to monitor the performance indicators of various rotating machinery, including fans, compressors, and mixers. It can enable the prediction of failures and the management of health parameters.	It has a wide range of applications and can design personalized fault diagnosis models according to the characteristics of different machines. It is often used for health monitoring and fault warning of large equipment.	- Vibration monitoring - Temperature and pressure monitoring - Data mining and pattern recognition - Multi-sensor fusion

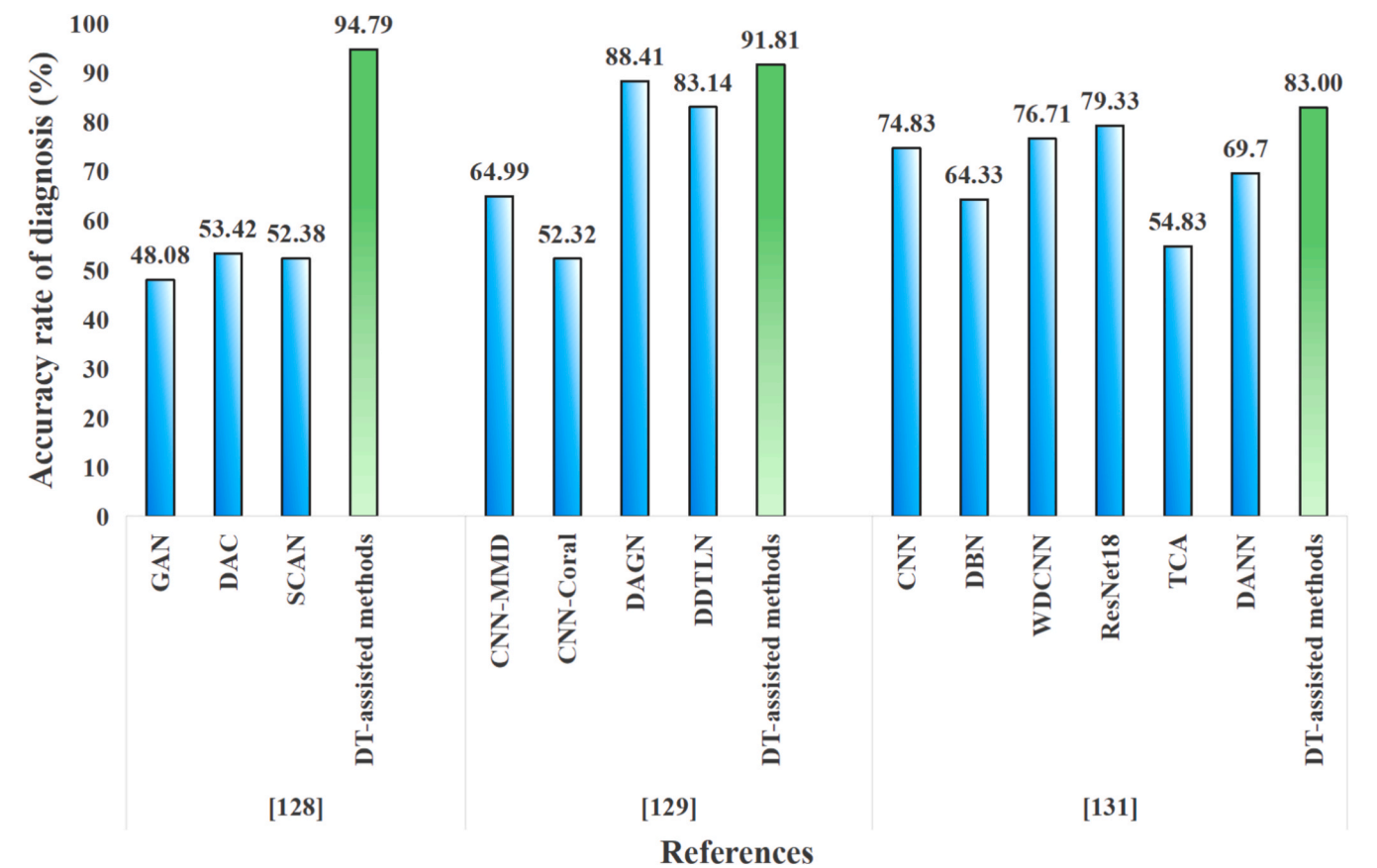


Fig. 16. Comparison of diagnostic accuracy.

collected from diverse industrial assets, thereby addressing the adaptive requirements of fault diagnosis. Zhang et al. [168] established a DT model of the gear transmission system and employed an improved multi-objective sparrow search algorithm to update it and obtain a sufficient

number of simulated fault samples. Jia et al. [169] used the hidden Markov model to describe the dynamic rules of the system, and constructed the mapping relationship between external excitation and monitoring signals to achieve high-precision fault diagnosis under

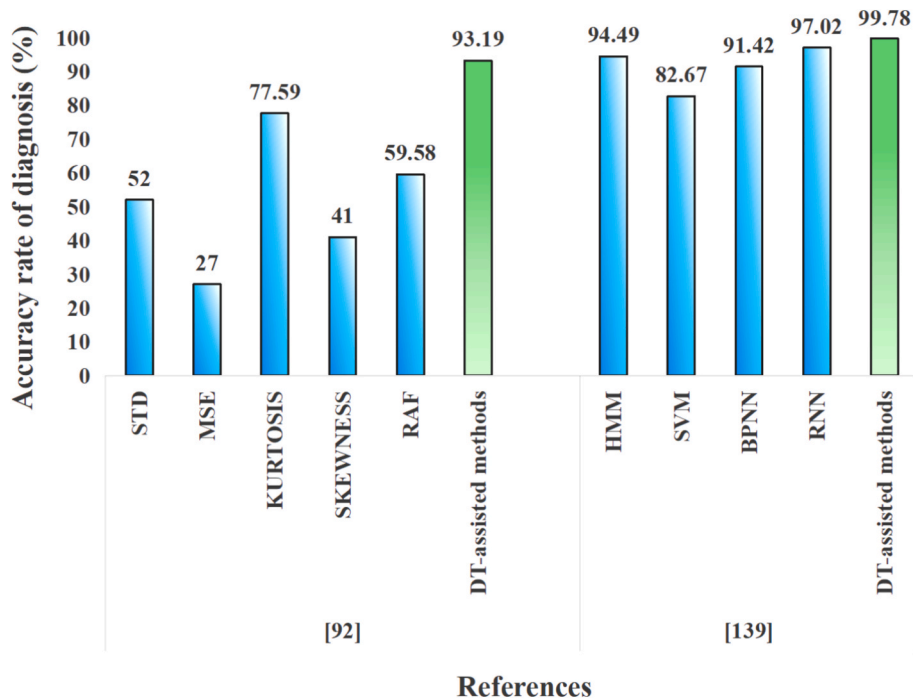


Fig. 17. Comparison of diagnostic accuracy.

dynamic conditions. Fig. 17 compares the diagnostic accuracy of various gear fault diagnosis methods as reported in different studies. As shown in Fig. 17, the DT-assisted method achieves significantly higher diagnostic accuracy in all the comparison cases, with peak values of 93.19 % and 99.78 %, clearly outperforming the traditional method. DT technology, through multi-body dynamics modeling, fusion of virtual and real signals, and generation of high-fidelity virtual samples, markedly improves the recognition accuracy of both minor and compound faults in gear transmissions, outperforming traditional diagnostic methods. Future work should focus on deeper integration of DT with advanced intelligent algorithms and refinement of model update mechanisms to meet industrial demands for precise diagnosis and early fault warning under complex operating conditions.

6.3. Motor

As the core part of rotating machinery, the normal operation of the motor is important to the stability and safety of mechanical systems. However, in the complex and changeable industrial environment, motors often face the risk of faults caused by multiple factors such as electrical, mechanical, thermal, operational, maintenance-related, and the environmental factors, which poses great challenges for accurate and timely fault diagnosis. By establishing a high-fidelity virtual model of the motor, integrating real-time operation data and historical monitoring data, and adopting multi-physics field simulation and intelligent algorithms, DT can accurately reproduce the dynamic behavior and internal state of the motor, thereby achieving early identification and prediction of faults.

Lee et al. [88] proposed a high-fidelity multi-physics model of a permanent magnet motor for generating fault data. Singh et al. [89] established the DT of squirrel-cage induction motors by using data-driven modeling and multiple physical principles. Hu et al. [170] proposed a DT architecture called Ramanujan DT to study the signal processing and intelligent diagnostic monitoring of induction motors based on the analysis of motor current characteristics. Xia et al. [171] established a DT model of an asynchronous motor based on multi-physics simulation, and proposed a semi-supervised motor fault diagnosis framework enhanced by DT. Lopes et al. [172] constructed a three-

dimensional FEM model of an induction motor incorporating rotor bar breakage faults and verified the influence of these faults by analyzing the current characteristics. Fig. 18 compares the diagnostic accuracy of various motor fault diagnosis methods as reported in different studies. As shown in Fig. 18, the DT-assisted method achieves significantly higher diagnostic accuracy in all the comparison cases, with peak values of 99.99 % and 92.89 %, clearly outperforming the traditional method. The above studies show that DT technology enhances motor fault diagnosis by integrating multi-physics modeling with data-driven and physical principles, leading to significant improvements in diagnostic accuracy. Future work should focus on advancing the integration of DT with high-fidelity simulations and multi-dimensional data analytics, as well as optimizing models for diverse motor types to enable early fault prediction and efficient maintenance, thereby driving the intelligent and precise evolution of motor operation and maintenance systems.

6.4. Pump

Pumps play a crucial role in mechanical equipment by performing functions such as fluid transportation, energy conversion, and pressure regulation. They are among the most widely used and critical devices in rotating machinery. The operating status of pumps directly affects the production efficiency and reliability of the entire system. The introduction of DT offers a new perspective and method for the fault diagnosis of pumps.

Chen et al. [131] proposed a fault diagnosis method for a three-cylinder pump by combining DT modeling and AdaBoost algorithm. Xu et al. [152] put forward a fault diagnosis method of centrifugal pumps, which utilized a DT framework powered by a graph transfer learning (GTL) model to solve the problem of insufficient fault data. To address the fault diagnosis problem of three-cylinder pumps, Liu et al. [153] constructed a DT-based fault diagnosis model by using TL method, which effectively improves the accuracy and reliability of diagnosis. Wang et al. [173] further built an IFD framework based on DT technology and improved the training performance of the ML model by using source domain data generated by DT. Building upon this work, Cai et al. [62] proposed a denoising autoencoder method combined with DT-assisted fault diagnosis, which further enhanced fault identification

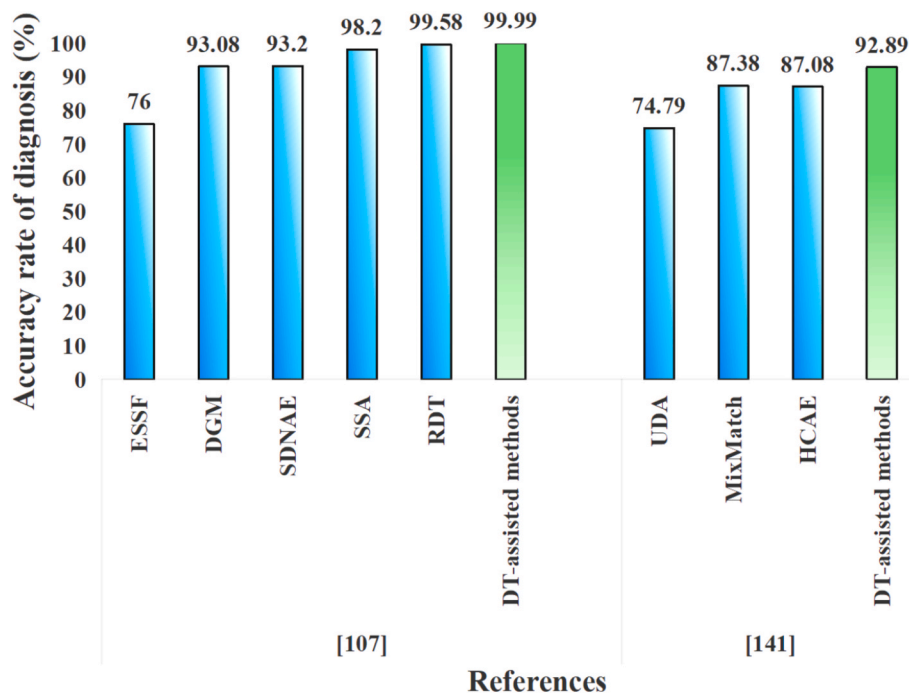


Fig. 18. Comparison of diagnostic accuracy.

performance for three-cylinder pumps and verified the potential of DT in supporting intelligent diagnosis under complex working conditions. Fig. 19 compares the diagnostic accuracy of various pump fault diagnosis methods as reported in different studies. As shown in Fig. 19, the DT-assisted method achieves significantly higher diagnostic accuracy in all the comparison cases, reaching peak accuracies of 95 %, 89.02 %, and 98.78 %, which clearly outperforms the traditional method. These findings indicate that DT technology in pump fault diagnosis effectively addresses issues such as insufficient fault data and low diagnostic accuracy under complex working conditions through integration with algorithms such as AdaBoost, GTL, and denoising autoencoders, thereby significantly enhancing diagnostic accuracy. In the future, collaboration between DT, multimodal data fusion, and adaptive learning algorithms can be further strengthened. Virtual model construction can be optimized according to the characteristics of different pump types to better meet the practical requirements for efficient operation and maintenance of pump equipment in industrial settings.

6.5. Turbine

Turbines serve as the core power component of aero-engines, gas turbines, wind turbines, hydraulic turbines, and nuclear power plants, and their operational status directly affects the safety and energy efficiency of these systems. DT technology has brought revolutionary advancements to fault diagnosis in rotating machinery, particularly demonstrating unique advantages in handling complex systems characterized by high temperatures, high pressures, and high rotational speeds.

Mousavi et al. [92] proposed a DT-based structural damage detection method, which utilized the data generated by the floating wind turbine model to evaluate the performance of this method. Liu et al. [107] built a DT model of a wind turbine drive system to monitor and simulate the actual operating conditions in real time. Dinh et al. [144] designed and implemented a DT model for a 2 MW-level doubly-fed wind power generation system. Li and Shen [150] proposed a DT-driven sensing approach, which extends the physical sensor platform into a virtual

sensor array to identify faulty sensors and adapt them by using appropriately estimated data. Fig. 20 compares the diagnostic accuracy of various turbine fault diagnosis methods as reported in different studies. As shown in Fig. 20, the DT-assisted method achieves significantly higher diagnostic accuracy in all the comparison cases, with peak values of 88 % and 99.1 %, clearly outperforming the traditional method. The above demonstrates that DT technology offers significant advantages in turbine fault diagnosis by constructing precise virtual models, enabling real-time status monitoring and data interaction, and expanding sensing capabilities. DT-assisted fault diagnosis methods achieve accuracy far exceeding traditional approaches. Future efforts should focus on advancing the deep integration of DT with multi-source data and intelligent algorithms to enhance the timeliness and reliability of turbine fault diagnosis, thereby providing stronger technical support for the efficient and safe operation and maintenance of equipment.

6.6. Other application scenarios

High-fidelity DT models of structures such as rolling bearings [159,160], gear transmissions [123,166], and motors [88,170] are constructed using FEM [172,174] and dynamic modeling methods [175,176], which are employed to simulate different various operating conditions and fault types, and generating realistic simulation data. The simulation data are then integrated with the actual data to augment the fault datasets and enhance the accuracy of data-driven fault diagnosis technology. By combining various modeling methods [132,177], visualization technologies [99,106], and Unity 3D software [107,178], this approach facilitates interaction between virtual and physical models, enables real-time monitoring of equipment operations, and provides essential support for maintenance personnel.

Additionally, DT has other applications in fault diagnosis and is also regarded as playing an important role in various diagnostic environments, such as photovoltaic systems [179–182], electric power systems [183–185], hydraulic systems [41,186,187], construction industries [188–192], the industrial internet of things [193–195], nuclear engineering [196,197], and other fields.

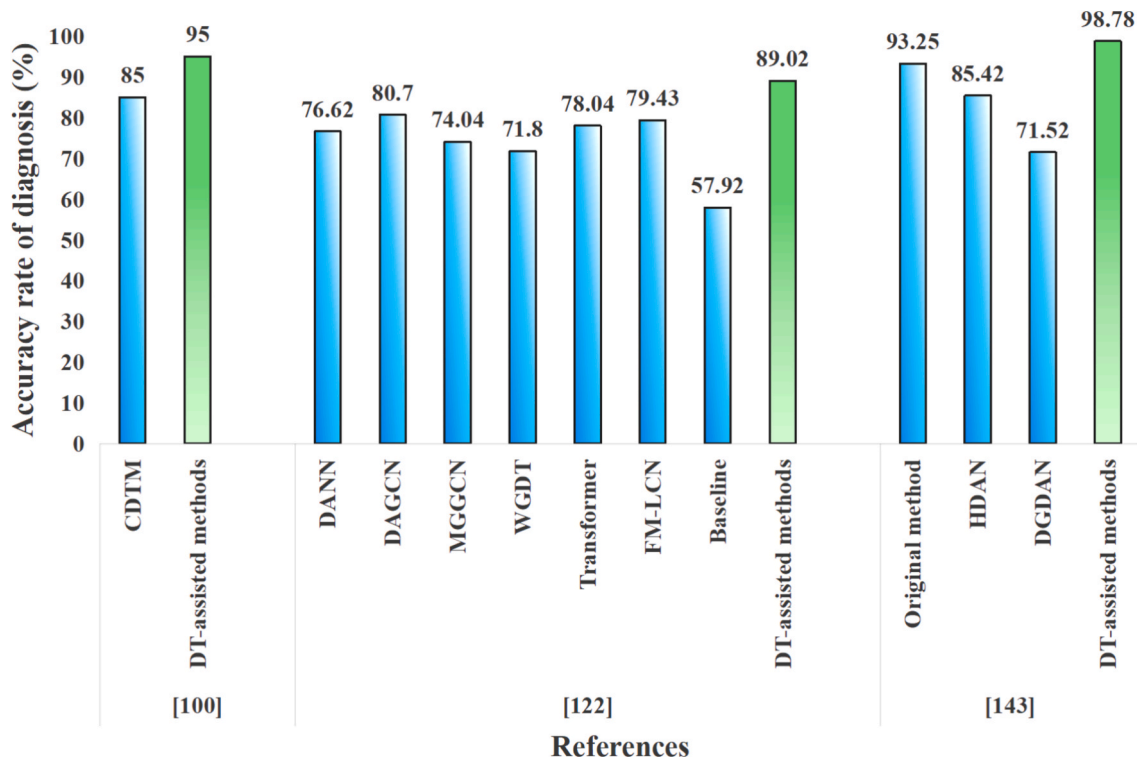


Fig. 19. Comparison of diagnostic accuracy.

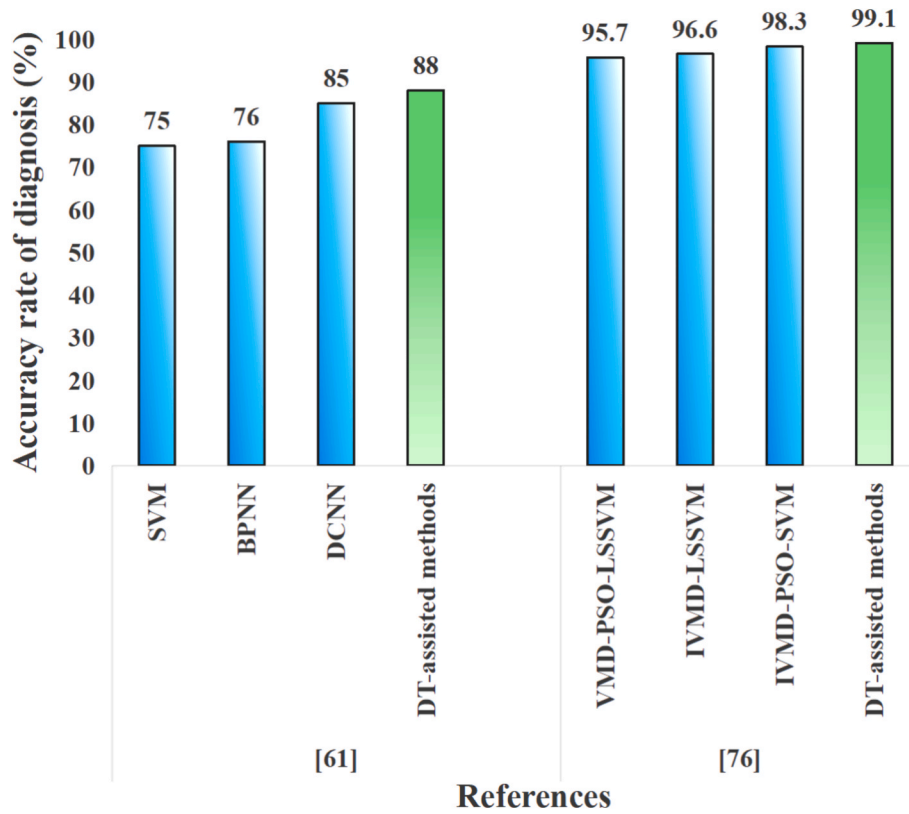


Fig. 20. Comparison of diagnostic accuracy.

7. Challenges and prospects for future development

Research on fault diagnosis integrated with DT technology is still in its infancy and is commonly referred to as “DT-assisted fault diagnosis” or “DT-driven fault diagnosis”. Although many challenges have been encountered in this field, there are also important opportunities. By systematically addressing these challenges, the use of DT will lead to substantial progress in fault diagnosis in the future.

7.1. Challenges

Currently, the application of Digital Twins (DTs) in rotating machinery fault diagnosis lags behind their adoption in other fields. Existing research primarily focuses on improving model accuracy and enabling real-time condition monitoring. The former generates the simulation data through high-fidelity digital models to support ML and DL-based diagnostic methods, while the latter focuses mainly on model visualization. It does not achieve true virtual-real interaction. Therefore, DT-assisted fault diagnosis and predictive maintenance remain under-developed in this domain.

(1) Difficulties in constructing high-fidelity digital models

The process of fault diagnosis using DTs is fundamentally dependent on high-precision models, which include geometric, physical, behavioral, and rule-based models. It is important to emphasize that physical entities are affected by numerous complex factors during operation, resulting in embedding nonlinear and dynamic characteristics into the fault diagnosis system. In addition, the occurrence of faults is often random and unpredictable, which greatly increases the complexity of high-fidelity digital model development. Take rolling bearings as an example. The emergence of compound faults accelerates the degradation of their performance, thereby posing a significant threat to the safe operation of equipment. Consequently, it is essential for the DT model to

possess the capability to precisely map the internal performance degradation state of rolling bearings and effectively uncover symptom characteristics such as randomness, secondary effects, coupling, and concealment that manifest during the diagnostic process. However, the complex motion and contact relationship among its internal components make it quite challenging to construct a high-fidelity dynamic model of compound faults.

(2) Low-quality fault data

DL is widely applied in rotating machinery fault diagnosis. The accuracy of this fault diagnosis method depends on the availability of high-quality data, including real-time operational data and historical fault data. However, in complex industrial environments, the operation of equipment may be disturbed by factors such as noise, resulting in problems such as incomplete or missing data. This makes it difficult to train a deep model to achieve accurate classification, and the diagnostic performance of the model deteriorates significantly. The construction of DT models has been significantly impeded by challenges such as limited sample sizes and data imbalance. Furthermore, the simulation data generated by the virtual models are influenced by model and computational accuracy, resulting in differences between the simulation data and the actual data, thereby affecting the accuracy of fault diagnosis.

(3) Limitations in computing and processing capabilities

Most current studies adopt dynamic methods and the FEM to construct DT models. However, due to the limitations of hardware and software, solving dynamic problems and FEMs often requires a significant amount of computing time. On the one hand, the development of high-fidelity digital models requires careful consideration of the structural complexity and multi-physics coupling characteristics of the models, such as the coupling of thermal-fluid-solid-magnetic multi-physics fields, which inevitably increases the simulation time. On the

other hand, to meet the real-time requirements of DT, methods such as model order reduction and dynamic model simplification are utilized to reduce the simulation time, which in turn leads to differences in the accuracy between the simulation data and the measured data. Therefore, balancing the accuracy of the model and the computational capability has become a major challenge for DT in the current application of fault diagnosis. DTs need to strike a dynamic and adaptable balance between fast computation and high accuracy.

Given the slow progress in hardware and software improvements, model lightweighting has emerged as a critical research issue in DT, as it can effectively enhance computational efficiency and significantly reduce solution time. However, this approach often results in a degradation of model accuracy. It performs well on specific architectures (e.g., ResNet, BERT) but shows limited generalization to other architectures or tasks. Most current lightweighting processes rely on manual expertise or hyperparameter tuning, lacking efficient automated solutions. Furthermore, model lightweighting can alter intrinsic feature representations, thereby negatively impacting the model's adversarial robustness.

(4) Lack of universal applicability

DT models generally face a core challenge: the lack of universal applicability. Existing DT models are usually deeply customized based on specific equipment, specific working conditions and specific fault modes. Therefore, the existing DT models are difficult to seamlessly migrate and reuse across different equipment, working conditions, or environments. This feature of high customization and low universality greatly increases the development cost and deployment difficulty of the DT system. Second, data-driven fault diagnosis methods depend on extensive datasets collected under specific operating conditions. When encountering new equipment or different working conditions, these models demonstrate limited generalization capability, frequently necessitating retraining or parameter recalibration.

Currently, IFD methods based on ML and DL are proliferating and have achieved remarkable accuracy in experimental settings. However, in real-world industrial applications, these models frequently suffer from severe performance degradation due to insufficient generalization capability. The core challenge stems from the inevitable distributional shift between training data and real-world deployment conditions, significantly impairing model generalization. Practical data collection is further compromised by sample selection bias, label noise, and class imbalance, all of which substantially degrade model generalizability. Moreover, there are often trade-offs between model lightweight and compression for edge deployment and generalization performance of the model. Therefore, developing a universally applicable fault diagnosis method based on deep analysis is a major challenge.

(5) Difficulties in real-time condition monitoring

DT-based real-time condition monitoring technology, while demonstrating significant potential for industrial health management, faces critical challenges in practical implementation. The inherent dynamic response latency frequently falls short of meeting the stringent real-time requirements of high-speed rotating machinery. Furthermore, reduced-order models suffer from accuracy degradation that compromises high-frequency dynamic characteristics, ultimately diminishing early weak fault detection capability. Another persistent issue stems from substantial discrepancies between sensor measurements and simulation outputs, creating problematic virtual-real data distribution mismatches. Moving forward, research efforts must focus on enhancing computational efficiency without sacrificing model accuracy to address these fundamental limitations.

(6) Challenges in security and privacy

Real physical entities are mirrored in virtual models through data

transmission, enabling equipment status monitoring and real-time fault diagnosis. DT-based fault diagnosis requires managing a large amount of operational data and sensitive information. In the process of data collection, storage and processing, the security and privacy of the system must be guaranteed. The collected data may contain critical information, ranging from core technologies to business secrets, and any leakage could result in significant organizational losses. Therefore, ensuring the security and confidentiality of fault diagnosis systems integrated with DT is a key challenge for future development.

7.2. Prospects for future development

Although DT technology has shown great application potential in the fault diagnosis of rotating machinery, it still faces several pressing challenges, such as the construction of high-fidelity digital models, the quality of twin data, the generalizability of DT methods, and limited computational capabilities. In response to these problems, there is an urgent need for continuous and in-depth research at multiple levels, from theoretical innovation and method improvement to engineering practice. To further advance the efficient application of DT technology in rotating machinery, future research directions and emerging development trends deserve in-depth exploration and analysis.

(1) Improvement of model accuracy

Physical processes at different scales often exhibit complex and heterogeneous dynamic behaviors in rotating machinery fault diagnosis, posing significant challenges to high-precision modeling and simulation. The discontinuity and interaction between scales significantly increase the complexity of multi-scale modeling and multi-physical field simulation, making it difficult for traditional single-scale or single-physical field modeling methods to accurately describe the global characteristics of the system. To address this problem, efforts should focus on developing modeling methods that integrate multi-scale and multi-physical field coupling mechanisms, and constructing high-fidelity DT models that can accurately represent the conditions of equipment across scales and physical domains. In addition, with the continuous advancement of modeling technology, advanced sensor technology, and data processing and fusion algorithms, the accuracy of DT models is expected to be further improved.

(2) Improvement of data quality

With the advancement of sensor technologies, increasing attention is being paid to the integration of multi-source data from various sensors and devices in the context of DT-based fault diagnosis. Through the fusion of diverse data sources, more accurate DT models can be constructed. These high-fidelity digital models are expected to generate simulation data closer to actual measurement data, thereby providing more realistic simulations and enhancing the overall quality across various fault types and operating conditions. Moreover, the thorough analysis of large volumes of historical and real-time data facilitates the identification of latent fault patterns, improving the quality of the measured fault data.

(3) Improvement of computing and processing capabilities

Advancements in computer hardware will bring substantial enhancements to computing and processing capabilities. Concurrently, cloud computing platforms and edge computing technologies offer new approaches for data processing, providing significant storage capacity and powerful computational resources that facilitate efficient simulation and training of DT models. Through cloud platform management, the DT model is utilized to perform unified diagnosis and analysis of data collected from various devices and regions. High-performance computing and distributed computing architectures support the

computing demands of large-scale data processing and complex modeling. Cloud computing and edge computing technologies can optimize the allocation and utilization of computing resources for DTs.

Future lightweighting approaches will evolve beyond single-technique applications toward integrated methodologies combining multi-modal and multi-task joint compression, thereby eliminating redundant model deployment overhead. These adaptive systems will dynamically reconfigure their architectures or computational paths according to input complexity and available resources, making them particularly well-suited for scenarios with stringent real-time computing requirements. Furthermore, lightweight models will establish symbiotic relationships with large foundation models (e.g., Transformers, GPT, and Diffusion Models), enabling collaborative inference across edge-cloud architectures.

(4) Improvement of the generalization performance of DT models

Model generalization represents a critical challenge in fault diagnosis research. Conventional fault diagnosis approaches relying on extensive labeled data are susceptible to overfitting. Emerging techniques such as self-supervised learning (SSL) and few-shot learning (FSL) address this by exploiting intrinsic data structural information to enhance model generalizability. Meta-learning further improves algorithm transferability, strengthening model adaptability across novel tasks and scenarios. Additionally, advancing multi-source/cross-domain generalization research is essential to overcome model dependence on training distribution characteristics, thereby enabling universal representation capability across diverse domains and tasks. Ultimately, generalization performance and model lightweighting are jointly optimized to produce lightweight models without compromising generalization capability.

(5) Standardization development

DTs have demonstrated powerful modeling and prediction capabilities in fault diagnosis. However, due to the imperfect standard system and the difficulty in cross-scenario migration of models, their large-scale engineering applications have been limited. To ensure the interconnection and interoperability of DTs among different manufacturers and platforms, it is necessary to establish a unified data format, a communication protocol, and an interface specification. A recent industry white paper has proposed a preliminary framework, which is conducive to guiding the modular development and deployment of DTs. By establishing a unified standardized system and adopting technologies such as modular design, TL, and proxy models, the interoperability and universality of DTs in rotating machinery fault diagnosis can be significantly enhanced. This enables the DT model to rapidly adapt to multi-device and multi-working condition environments.

(6) Development of intelligent fault diagnosis

In recent years, IFD methods for mechanical faults have made remarkable progress in dealing with data imbalance, open-set identification, small-sample transfer, and dynamic working-condition adaptation. For instance, fault identification models based on imbalance learning and open-set generalization mechanisms have achieved active identification and risk assessment of unknown faults. The *meta-learning* framework, designed to enhance the mechanical diagnostic capability, improves the transferability and reliability of the diagnostic model and constructs a *meta-dataset* of fault tasks to train a general task-adaptability model [198–200]. The lightweight progressive joint transfer ensemble network method, inspired by Markov processes, addresses the challenges of imbalance and dynamic adaptation, enables robust fault identification in unbalanced data and non-stationary environments, and significantly improves diagnostic accuracy and model

stability [201–204].

In the future, IFD is expected to shift from traditional rule-based and shallow learning methods to DL and large model-driven approaches. Large-scale pre-trained models will be closely integrated with DTs, enabling the extraction of common fault features from massive cross-domain data to effectively mitigate issues such as small sample sizes and distribution shifts. Furthermore, lightweight fine-tuning and online learning mechanisms will enable efficient adaptive diagnosis for specific devices. Ultimately, IFD will be driven by large models and deeply integrated with DTs to achieve end-to-end, highly reliable, and multi-task collaborative diagnosis from the equipment level and working condition level to the system level.

(7) Enhancing the capacity for data security assurance

In the future, collaborative efforts will be required across multiple dimensions, including data privacy protection, prevention of sensitive information leakage during model training, large-scale distributed learning, and security of massive data. These efforts aim to establish a full-chain, secure, and trustworthy ecosystem for rotating machinery fault diagnosis based on DTs. On the one hand, it is necessary to identify and manage personal and enterprise-sensitive information in the training data of the DT model, and to adopt privacy-enhancing technologies such as differential privacy and homomorphic encryption for data desensitization and encryption. On the other hand, distributed collaborative mechanisms such as federated learning and blockchain can effectively minimize data exposure during the training and update DT model. Meanwhile, multi-factor authentication, role-based access control, and immutable audit logs should be implemented at every stage of data storage, transmission, and access to ensure protection against cyberattacks and compliance with the big data platform regulations.

8. Conclusions

Although the application of DT in IFD for rotating machinery remains nascent, it has gained rapid momentum and demonstrated significant potential in key sectors including wind power, aero-engines, and industrial motors. Nevertheless, existing research faces core challenges including insufficient model fidelity, low data fusion efficiency, and limited generalization ability, among others. In this paper, the latest progress in this field is reviewed, and novel perspectives are proposed. The main contributions are as follows:

- (1) It provides a comprehensive overview of the latest research advances and implementation methodologies of DT technology for rotating machinery fault diagnosis. Specifically, it covers the construction of high-fidelity digital models, acquisition of high-quality data, real-time condition monitoring, and visualization methods for fault diagnosis. These contributions are instrumental in improving the adaptability and effectiveness of DT applications in IFD.
- (2) The model construction paradigm, multi-source data fusion mechanisms, and engineering applications of DT technology for rotating machinery fault diagnosis are systematically examined. This study provides theoretical guidance and technical references for DT model optimization and subsequent research, thereby contributing significantly to advancing fault diagnosis technologies. Based on current research progress, future work will focus on developing hybrid modeling methods that integrate physical mechanisms with data-driven approaches, investigating advanced spatio-temporal alignment and feature extraction algorithms for multimodal heterogeneous data, and establishing standardized, scalable DT frameworks for fault diagnosis to facilitate technological translation and practical applications.
- (3) In conclusion, the key technical challenges currently hindering DT technology in rotating machinery fault diagnosis applications

are summarized, and the corresponding development strategies are proposed. Although significant bottlenecks persist in model fidelity, data fusion efficiency, and system generalization capability, the convergence of next-generation artificial intelligence algorithms, high-performance computing, and intelligent sensing technologies demonstrates promising application prospects of DT in this field.

In summary, the current application status of DT technology in IFD for rotating machinery is elucidated, emphasizing the need for sustained research innovation to fully realize the potential of DT. Although DT-based fault diagnosis research remains at an early stage of development, overcoming existing challenges through deep integration with neural network technologies will enable DT to deliver more reliable, efficient, and intelligent diagnostic solutions for rotating machinery.

CRedit authorship contribution statement

Jianxiong Wei: Writing – review & editing, Writing – original draft, Methodology. **Changfeng Yan:** Funding acquisition, Conceptualization. **Yaofeng Liu:** Formal analysis. **Bin Liu:** Supervision, Software. **Lixiao Wu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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